



Modelling Financial Markets

Chapter 18: Systemic Risk and Networks



Daniel Traian PELE

Bucharest University of Economic Studies

IDA Institute Digital Assets

Blockchain Research Center

AI4EFin Artificial Intelligence for Energy Finance

Romanian Academy, Institute for Economic Forecasting

MSCA Digital Finance

Acronyms used in this material (1/2)

- ▣ **AI:** Artificial Intelligence
- ▣ **BET:** Bucharest Exchange Trading (index)
- ▣ **BH:** Benjamini–Hochberg (procedure)
- ▣ **BIC:** Bayesian Information Criterion
- ▣ **BNP:** BNP Paribas SA (Euronext Paris ticker; Banque Nationale de Paris)
- ▣ **BVB:** Bursa de Valori București — *the Bucharest Stock Exchange*
- ▣ **CATFIN:** CATastrophic risk in the FINancial sector
- ▣ **CDS:** Credit Default Swap
- ▣ **CI:** Confidence Interval
- ▣ **COVID-19:** COronaVirus Disease 2019
- ▣ **DCC:** Dynamic Conditional Correlation
- ▣ **EBA:** European Banking Authority
- ▣ **ECB:** European Central Bank
- ▣ **ES:** Expected Shortfall
- ▣ **EU:** European Union
- ▣ **FDIC:** Federal Deposit Insurance Corporation
- ▣ **FDR:** False Discovery Rate
- ▣ **FEVD:** Forecast Error Variance Decomposition
- ▣ **FRM:** Financial Risk Meter
- ▣ **GAAP:** Generally Accepted Accounting Principles (US)
- ▣ **GACV:** Generalized Approximate Cross-Validation
- ▣ **GARCH:** Generalised AutoRegressive Conditional Heteroskedasticity

Acronyms used in this material (2/2)

- ▣ **GJR**: Glosten–Jagannathan–Runkle (asymmetric GARCH)
- ▣ **HAC**: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- ▣ **HSBC**: HSBC Holdings plc (originally the Hongkong and Shanghai Banking Corporation)
- ▣ **IFRS**: International Financial Reporting Standards
- ▣ **ING**: ING Groep (Internationale Nederlanden Groep) — *the Dutch banking group ING*
- ▣ **LASSO**: Least Absolute Shrinkage and Selection Operator
- ▣ **LRMES**: Long-Run Marginal Expected Shortfall
- ▣ **MES**: Marginal Expected Shortfall
- ▣ **OLS**: Ordinary Least Squares
- ▣ **QFVAR**: Quantile Factor Vector AutoRegression
- ▣ **QML**: Quasi-Maximum Likelihood
- ▣ **RTD**: Relative Tail Dependence
- ▣ **SE**: Standard Error
- ▣ **SG**: Société Générale — *the French banking group Société Générale*
- ▣ **SRISK**: Systemic RISK (capital shortfall measure)
- ▣ **SVB**: Silicon Valley Bank
- ▣ **TENET**: Tail-Event driven NETwork risk
- ▣ **US**: United States
- ▣ **VAR**: Vector AutoRegression
- ▣ **VIX**: CBOE Volatility Index

Systemic Risk and the Route of This Chapter

From one bank to the system

- **Systemic risk:** the risk that the distress of one or several institutions impairs the financial system as a whole, and through it the real economy
- An **externality:** a bank does not pay for the losses its failure imposes on others
- Three families of measures Billio, Getmansky, Lo & Pelizzon (2012); survey Benoit, Colliard, Hurlin & Pérignon (2017)
 - ▶ **Exposure:** what does the bank lose when the market crashes? MES, SRISK
 - ▶ **Contribution:** how much worse is the system when the bank is in distress? ΔCoVaR (CoVaR: the Value-at-Risk, VaR, of the system conditional on the bank)
 - ▶ **Connectedness:** through which links do shocks travel? Spillover and tail networks

Route of this chapter

- Part I: data and conventions; exposure (MES, LRMEs, SRISK); contribution (CoVaR, ΔCoVaR) and how far rankings can be trusted
- Part II: spillover networks, the Financial Risk Meter, stress tests and climate risk, a case study on quantile connectedness, one pass of the AI discovery loop
- Proofs, estimation details and robustness checks sit in the **Appendix** at the end
- Textbooks: Franke, Härdle & Hafner (2019), Härdle, Chen & Overbeck (2017)

Bank Runs: Northern Rock 2007 and Silicon Valley Bank 2023



Depositors queue outside a Northern Rock branch,
Brighton, 14 September 2007

Photo: [Dominic Alves \(2007\)](#); CC BY 2.0; [Wikimedia](#)

[Commons](#)

- Northern Rock (2007): a British mortgage lender hit by the freeze in funding markets after the US subprime losses; depositors queued to withdraw their savings
- SVB (2023): a digital run; the FDIC took the bank over on 10 March [FDIC \(2023\)](#); two days later all depositors of SVB and Signature Bank were protected on **systemic** risk grounds [Federal Reserve Board \(2023\)](#)
 - ▶ The stress-testing section measures the loss of a bank portfolio in that window
- Channels of contagion
 - ▶ **Funding runs**: depositors and lenders withdraw from any bank that looks similar
 - ▶ **Common exposures** and **fire sales**: forced selling lowers prices for every holder
 - ▶ **Counterparty links**: one bank's liability is another bank's asset, so a default passes losses to its creditors

Data and Conventions (1/2): Banks and Prices

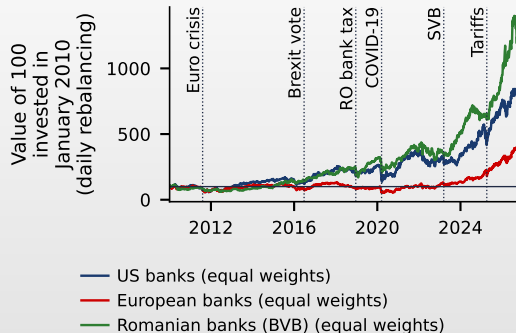
Group	Series	Market index
US banks	JPMorgan, Bank of America, Citigroup, Goldman Sachs, Morgan Stanley, Wells Fargo	S&P 500
European banks	Deutsche Bank, BNP Paribas, Santander, ING, HSBC	Euro Stoxx 50
Romanian banks	Banca Transilvania, BRD-Groupe Société Générale	BET
Volatility	the VIX index (implied volatility of the S&P 500)	

- Daily prices, January 2010 to 18 September 2026, adjusted for dividends and splits
 - ▶ Aligned on 3 960 common trading days first, then daily log returns $r_t = 100(\ln P_t - \ln P_{t-1})$, in %
- Special cases
 - ▶ Romanian banks: days without trades removed
 - ▶ Historical stress scenarios for the international banks start in 2007

Data and Conventions (2/2): Signs and the System

- ▣ Return X , loss $L = -X$; the level α is a **tail probability** ([Chapter 7](#))
 - ▶ We write VaR 1%, CoVaR 1%, MES 5%
 - ▶ $\text{VaR}_\alpha(X) = -q_\alpha(X)$, with q_α the α -quantile of X
 - ▶ VaR, CoVaR and MES are positive numbers when they are losses
- ▣ **System** return X^{sys} : the equal-weighted portfolio of the other banks of the same region
 - ▶ $X^{\text{sys}} = \frac{1}{n} \sum_j X_j$, the average daily log return of the n other banks
 - ▶ A linear aggregate: a first-order approximation of the log return of the daily rebalanced portfolio
 - ▶ MES, CoVaR and the Euler identity are stated for this aggregate

Three Banking Systems, 2010–2026

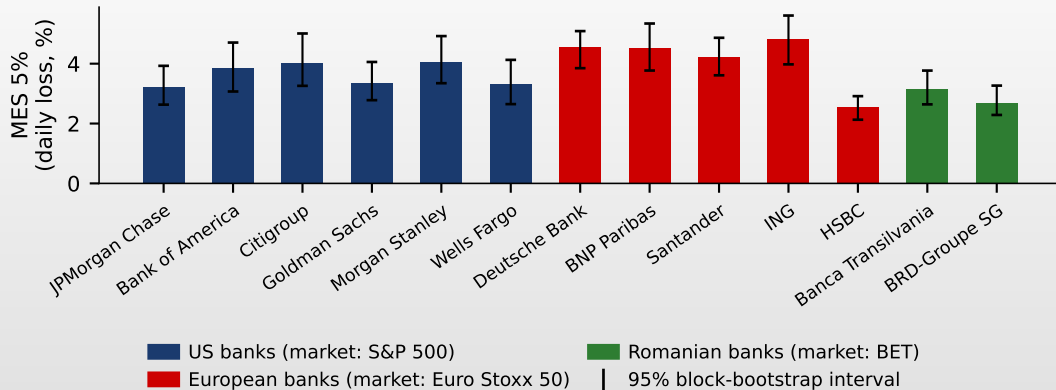


- Equal-weighted portfolios of the banks of each region, rebalanced daily
 - ▶ Vertical lines mark the episodes used in this chapter
- 100 invested in January 2010 became 791 (US), 384 (Europe), 1 233 (Romania)
 - ▶ Maximum drawdown: US 51% (2011-11-23), Europe 61% (2020-04-21), Romania 47% (2012-06-06)
 - ▶ Annualised volatility: 29%, 30%, 23%; indices: 18%, 21%, 18%
- The falls are **simultaneous**: euro crisis, COVID-19, 2023 bank failures
 - ▶ Romanian banks also have their own shock: the bank-tax announcement of December 2018

Marginal Expected Shortfall and the Euler Property

- **MES** (Marginal Expected Shortfall) of bank i at level α Acharya, Pedersen, Philippon & Richardson (2017)
 - ▶ $MES_{i,\alpha} = -\mathbb{E}[X_i \mid X_m \leq q_\alpha(X_m)]$
 - ▶ The average loss of the bank on the worst α share of days for the market m
- Why “marginal”: if the conditioning variable is the portfolio itself, $m = \sum_i w_i X_i$, then $ES_\alpha(X_m) = \sum_i w_i MES_{i,\alpha}$ (Euler allocation; proof in the [Appendix](#))
 - ▶ w_i : portfolio weights; $ES_\alpha(X_m) = -\mathbb{E}[X_m \mid X_m \leq q_\alpha(X_m)]$: the average loss of m on its own worst α days
 - ▶ Our markets (S&P 500, Euro Stoxx 50, BET) are **not** portfolios of these 13 banks: here MES measures **exposure**, not a share of the index ES
- Check on our data (six US banks, $\alpha = 5\%$, 198 tail days): portfolio $ES_{5\%} = 4.32\% =$ average MES conditional on the portfolio (4.32%); conditioning on the S&P 500 instead gives 3.63%
- Historical estimate: average $-X_i$ over the days with $X_m \leq \hat{q}_\alpha(X_m)$; with $\alpha = 5\%$ and 3960 days, about 200 days
- Market m : S&P 500 for US banks, Euro Stoxx 50 for European banks, BET for Romanian banks

MES 5% of Thirteen Banks (1/2)



MES 5% of Thirteen Banks (2/2)

- Average daily loss of each bank on the worst 5% of days for its market index, 2010–2026
 - ▶ 95% intervals: moving-block bootstrap, blocks of 20 days
- US banks: 3.22% (JPMorgan) to 4.04% (Morgan Stanley); Europe up to 4.81% (ING)
 - ▶ Same days, market: 2.72% (S&P 500), 3.16% (Euro Stoxx 50): banks lose more
- HSBC is the exception in Europe (2.54%, below the index)
 - ▶ A London listing with business concentrated in Asia
- Banca Transilvania 3.14%, BRD 2.69%; BET itself 2.70%
- Overlapping marginal intervals are not a test of equality
 - ▶ Bank-to-bank differences need a paired test on joint bootstrap draws (Section 3)

From MES to a Crisis: LRMES

- **LRMES** (Long-Run MES): the expected fall of the bank's market equity in a crisis
 - ▶ $\text{LRMES}_{i,t} = -\mathbb{E}_t [R_{i,t+1:t+h} \mid R_{m,t+1:t+h} < C]$, crisis: a market fall below C over h days
 - ▶ $R_{i,t+1:t+h}$, $R_{m,t+1:t+h}$: arithmetic returns of bank i and of the market from $t+1$ to $t+h$
 - ▶ Acharya, Engle & Richardson (2012) (Section I): $C = -40\%$ over six months; Brownlees & Engle (2017) (Section 1.1): $C = -10\%$ over $h = 22$ days
- Shortcut of Acharya, Engle & Richardson (2012) (Section I), used when no simulation is available
 - ▶ $\text{LRMES}_i \approx 1 - \exp(-18 \times \text{MES}_i^{2\%})$
 - ▶ $\text{MES}_i^{2\%}$: average daily loss of the bank on days when the market falls by more than 2% (in decimals)
 - ▶ Example: $\text{MES}^{2\%} = 3\%$ gives $1 - e^{-0.54} = 41.7\%$; the 18 is a calibration constant (Seminar 18, A5)
- Shortcut on our banks: 37% (HSBC) to 58% (ING); JPMorgan 48%, Banca Transilvania 52%
 - ▶ Market falls of more than 2%: 138 days in the US, 196 in the euro area, 111 on the BVB

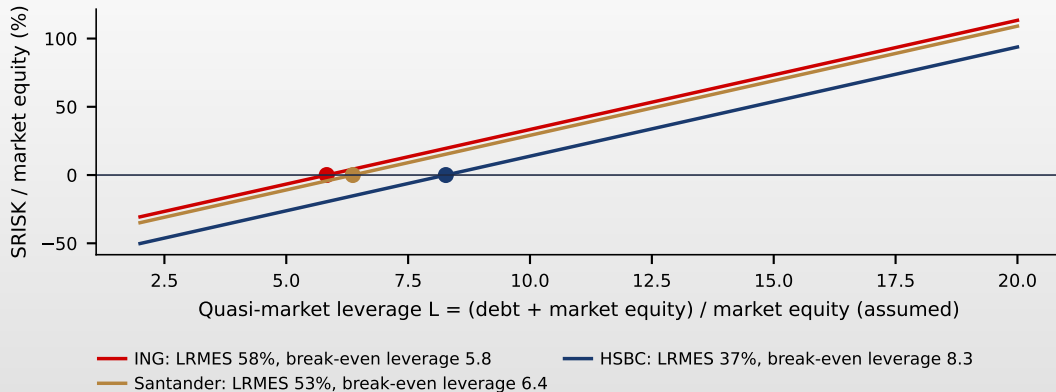
SRISK and the Break-Even Leverage (1/2)

- A bank must keep equity of at least a share k of its quasi-assets [Brownlees & Engle \(2017\)](#), Section 1.1, Eq. (1)
 - ▶ Quasi-assets: book debt D plus market equity W
 - ▶ $SRISK_i = k D_i - (1 - k)(1 - LRMES_i) W_i$, assuming debt is not renegotiated in the crisis
 - ▶ $k = 8\%$ for US firms [Brownlees & Engle \(2017\)](#); $k = 5.5\%$ for European firms (IFRS balance sheets report larger assets than US GAAP) [Engle, Jondeau & Rockinger \(2015\)](#)
- Positive SRISK = capital shortfall
 - ▶ SRISK of the system: the sum of the positive values
 - ▶ V-Lab, [New York University Stern School of Business](#) publishes it for large banks, from their balance sheets

SRISK and the Break-Even Leverage (2/2)

- Divide by W ; quasi-market leverage $L = (D + W)/W$
 - ▶ $\frac{\text{SRISK}}{W} = k(L - 1) - (1 - k)(1 - \text{LRMES})$
- SRISK is positive once leverage exceeds the **break-even leverage**
 - ▶ $L^* = 1 + \frac{(1 - k)(1 - \text{LRMES})}{k}$
- Example: $D = 900$, $W = 100$ (so $L = 10$), $\text{LRMES} = 41.7\%$
 - ▶ $\text{SRISK} = 0.08 \times 900 - 0.92 \times (1 - 41.7\%) \times 100 = 72 - 53.6 = 18.4$
 - ▶ $L^* = 7.7 < L = 10$: a shortfall
- Our data contain prices, not balance sheets
 - ▶ We report L^* from the estimated LRMES and treat leverage as an **assumption**

SRISK as a Function of Leverage (1/2)



SRISK as a Function of Leverage (2/2)

- Banks with the highest, median and lowest LRMES; dots mark L^*
 - ▶ Leverage on the horizontal axis is assumed, not observed
- With $k = 8\%$: $L^* = 5.8$ (ING), 6.4 (Santander), 8.3 (HSBC)
 - ▶ US: JPMorgan 7.0, Morgan Stanley 6.0; Romania: Banca Transilvania 6.5, BRD 7.1
- Debt above about 5–7 times market equity: new capital needed in a 40% market crash
- With $k = 5.5\%$ the thresholds rise (ING 8.2, HSBC 11.9)
 - ▶ The prudential ratio is a policy choice, and SRISK depends on it
- Limits: LRMES extrapolates daily data; market equity itself falls in a crisis

LRMES: GJR–GARCH, DCC and Simulation (1/2): the Model

- Model of [Brownlees & Engle \(2017\)](#), Section 1.2: zero-mean daily log returns of the bank and of the market
 - ▶ GJR–GARCH(1,1) volatilities [Glosten, Jagannathan & Runkle \(1993\)](#); GARCH: generalized autoregressive conditional heteroskedasticity
 - ▶ A DCC(1,1) correlation (DCC: dynamic conditional correlation) [Engle \(2002\)](#); two-step quasi-maximum likelihood (QML)
- Volatility, for the bank and for the market: today's variance reacts to yesterday's squared return, more after a fall
 - ▶ $\sigma_t^2 = \omega + (\alpha + \gamma 1\{r_{t-1} < 0\}) r_{t-1}^2 + \beta \sigma_{t-1}^2$
 - ▶ σ_t^2 : conditional variance of the return r_t ; $\omega > 0$: constant; α : reaction to a shock; $\gamma > 0$: extra reaction after a fall (leverage effect); β : persistence; $1\{\cdot\}$: 1 if the condition holds, else 0
- Correlation: a GARCH-type recursion for the matrix Q_t , rescaled to a correlation
 - ▶ $Q_t = (1 - a - b) S + a \epsilon_{t-1} \epsilon_{t-1}^\top + b Q_{t-1}$, $\rho_t = Q_{12,t} / \sqrt{Q_{11,t} Q_{22,t}}$
 - ▶ $\epsilon_t = (r_{it}/\sigma_{it}, r_{mt}/\sigma_{mt})^\top$: standardised returns; S : their sample correlation matrix; a : reaction, b : persistence of the correlation; ρ_t : today's bank–market correlation

LRMES: GJR–GARCH, DCC and Simulation (2/2): Simulation

- Simulation (Brownlees & Engle, ▶ **Appendix** A; details in the ▶ **Appendix** of this deck)
 - ▶ Resample **pairs** of standardised innovations (ξ_{it}, ϵ_{mt}) of bank and market
 - ▶ Filter h days ahead from today's σ^2 and ρ ; keep the paths with $R_{m,t+1:t+h} < C$
 - ▶ LRMES = minus the average bank return over these crisis paths
 - ▶ Resampling pairs keeps the tail dependence that the Normal distribution misses
- Data to 2026-09-18; paths per market
 - ▶ 200 000 paths of 22 days ($C = -10\%$); 400 000 paths of 126 days ($C = -40\%$)
- Probability of the 22-day crisis
 - ▶ Today: 1.3% (S&P 500), 4.3% (Euro Stoxx 50), 4.0% (BET)
 - ▶ On 31 March 2020: 12%, 19%, 8%

LRMES by Simulation: Thirteen Banks (1/2)

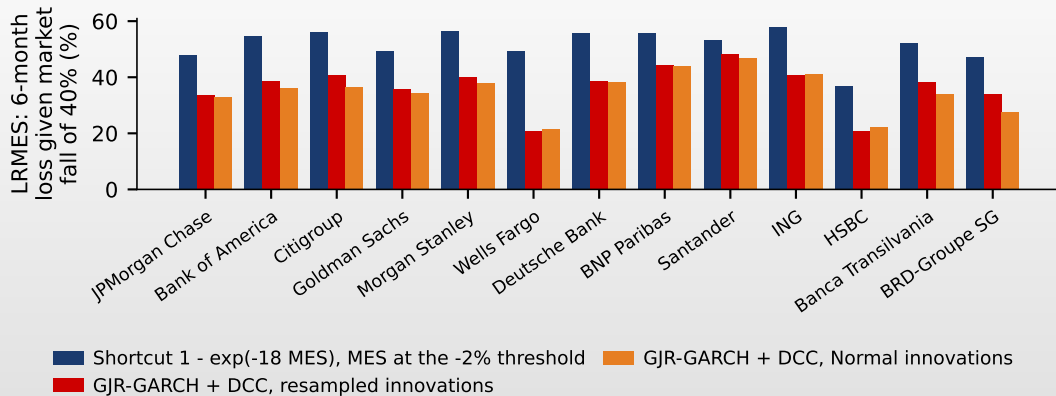
Bank	LRMES today	90% range	LRMES 31.03.2020	ρ_T	L^*
JPMorgan	10.6	[-2, 25]	23.7	0.53	11.3
Bank of America	14.3	[-4, 35]	23.2	0.44	10.9
Citigroup	13.0	[-1, 28]	32.8	0.54	11.0
Goldman Sachs	16.5	[1, 34]	25.0	0.65	10.6
Morgan Stanley	16.3	[1, 32]	27.2	0.65	10.6
Wells Fargo	6.7	[-8, 24]	22.9	0.32	11.7
Deutsche Bank	14.4	[2, 28]	25.7	0.66	10.8
BNP Paribas	15.6	[2, 33]	28.5	0.68	10.7
Santander	16.6	[3, 34]	26.4	0.76	10.6
ING	12.5	[1, 27]	32.9	0.68	11.1
HSBC	8.3	[-2, 19]	10.9	0.58	11.5
Banca Transilvania	12.4	[-2, 34]	17.2	0.74	11.1
BRD-Groupe SG	11.4	[-2, 28]	15.8	0.64	11.2

- LRMES (%): market fall below -10% in 22 days [Brownlees & Engle \(2017\)](#); 90% range: 5% and 95% quantiles of the bank's loss on the crisis paths; ρ_T : last DCC correlation; L^* with $k = 8\%$

LRMES by Simulation: Thirteen Banks (2/2)

- LRMES is **conditional**: it moves with today's volatility and correlation
 - ▶ JPMorgan: 10.6% today, 23.7% with the information of 31 March 2020
 - ▶ Deutsche Bank: 14.4% and 25.7%
- The 90% range is wide [Brownlees & Engle \(2017\)](#), Eq. (2)
 - ▶ LRMES is an average over crisis paths; the shortfall in one particular crisis can be much larger
- The scenario, not only the bank, sets the level
 - ▶ One-month, -10% crisis: LRMES about a third of the six-month, -40% numbers of [Acharya, Engle & Richardson \(2012\)](#)

Model Risk: LRMES under Three Specifications (1/2)



Model Risk: LRMES under Three Specifications (2/2)

- Six-month crisis, market fall of 40% [Acharya, Engle & Richardson \(2012\)](#); data to 2026-09-18
 - ▶ Shortcut $1 - e^{-18\text{MES}^2\%}$ vs GJR-GARCH + DCC simulation with resampled or Normal innovations
- Simulated LRMES over the shortcut: 0.42 to 0.90; rank correlation 0.77
 - ▶ JPMorgan: 48% (shortcut), 33% (resampled), 33% (Normal)
 - ▶ ING: 58%, 41%, 41%
- The shortcut uses the whole history; the simulation starts from today's volatility
- Resampled vs Normal innovations: rank correlation 0.95
 - ▶ The innovation distribution changes levels more than rankings
- Same data, different defensible models, different numbers: model risk [Danielsson, James, Valenzuela & Zer \(2016\)](#)

 MFM_ch18_srisk_sim

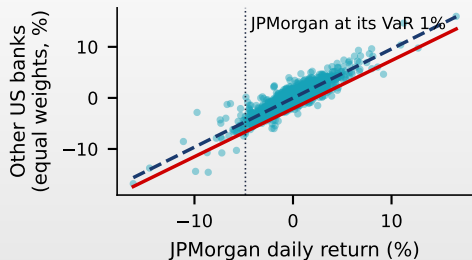
CoVaR: the VaR of the System Given a Bank in Distress

- **CoVaR** of the system conditional on bank i at its VaR [Adrian & Brunnermeier \(2016\)](#); X^{sys} , X_i : returns of the system and of bank i
 - ▶ $P\left(X^{sys} \leq -\text{CoVaR}_\alpha^{sys|i} \mid X_i = -\text{VaR}_{i,\alpha}\right) = \alpha$
 - ▶ The loss of the system that is exceeded with probability α on a day when bank i is at its own VaR
- **ΔCoVaR** : the change from the median state of the bank to its distress state
 - ▶ $\Delta\text{CoVaR}_\alpha^{sys|i} = \text{CoVaR}_\alpha^{sys|X_i=q_\alpha} - \text{CoVaR}_\alpha^{sys|X_i=q_{50\%}}$
 - ▶ $\text{CoVaR}_\alpha^{sys|X_i=q}$: CoVaR with the bank return fixed at its quantile q ; $q_{50\%}$: the median, a normal day
- ΔCoVaR measures the **contribution** of bank i to the tail risk of the system
- Reversing the roles gives the **exposure CoVaR**: the VaR of bank i when the system is in distress

Estimating CoVaR by Quantile Regression

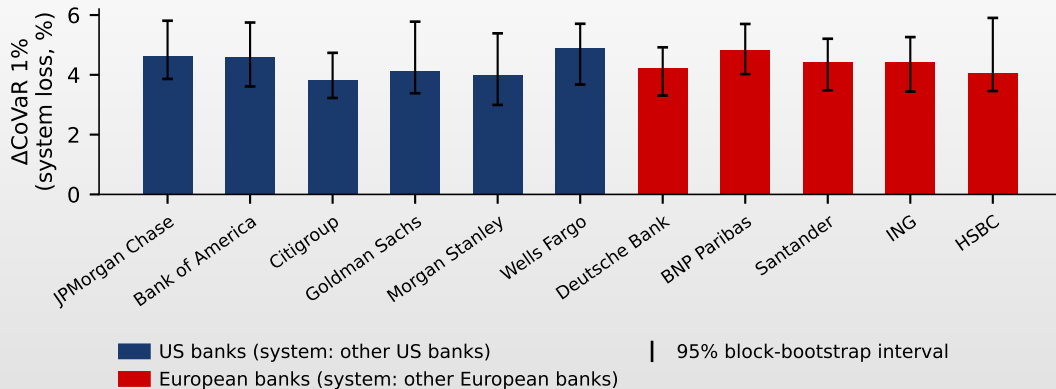
- The α -quantile of Y given x is linear: $q_\alpha(Y | x) = a_\alpha + b_\alpha x$ [Koenker & Bassett \(1978\)](#)
 - ▶ Here Y : the system return, x : the bank return; on the data y_t, x_t : $(\hat{a}_\alpha, \hat{b}_\alpha) = \arg \min_{a,b} \sum_t \rho_\alpha(y_t - a - bx_t)$,
 $\rho_\alpha(u) = u(\alpha - 1\{u < 0\})$
 - ▶ ρ_α is the **check function**: it weights negative residuals by $1 - \alpha$ and positive ones by α ; $\alpha = 50\%$ gives the median regression
- 1. Quantile regression of the system on the bank at level α
 - ▶ $\hat{q}_\alpha(X^{sys} | X_i) = \hat{a}_\alpha + \hat{b}_\alpha X_i$
- 2. Plug in the bank's own quantiles: $\hat{q}_\alpha(X_i)$ (distress) and $\hat{q}_{50\%}(X_i)$ (median)
 - ▶ $\text{CoVaR}_\alpha^{sys|i} = - \left(\hat{a}_\alpha + \hat{b}_\alpha \hat{q}_\alpha(X_i) \right)$
- 3. Take the difference
 - ▶ $\Delta \text{CoVaR}_\alpha^{sys|i} = \hat{b}_\alpha (\hat{q}_{50\%}(X_i) - \hat{q}_\alpha(X_i))$
 - ▶ Only the slope matters: the intercept cancels
- Uncertainty: moving-block bootstrap of the days (blocks of 20 days keep volatility clusters together), 999 resamples

Quantile Regression: JPMorgan and the Other US Banks



- Daily returns, 2010-2026
- 1% quantile regression: $-2.10 + 0.94 x$
- - 50% quantile regression: $-0.03 + 0.96 x$

- Daily returns 2010–2026, 1% and 50% quantile regression lines
 - ▶ Dotted line: JPMorgan at its VaR 1%
- 1% line: $\hat{q}_{1\%} = -2.10 + 0.94 X_i$; median line: $-0.03 + 0.96 X_i$
 - ▶ Similar slopes: the 1% line is shifted down, not steeper
- JPMorgan's VaR 1% is 4.83%; the other US banks then have CoVaR 1% = 6.64%
 - ▶ System VaR 1%: 5.07%; with JPMorgan at its median: 2.03%
- $\Delta \text{CoVaR}_{1\%} = 4.62\%$: JPMorgan in distress adds about 4.62 points
 - ▶ Association, not causation: a common shock hits all banks on the same day

ΔCoVaR 1% of Eleven Banks (1/2)

Δ CoVaR 1% of Eleven Banks (2/2)

- System: equal-weighted portfolio of the other banks of the same region
 - ▶ 95% moving-block bootstrap intervals
- From 3.81% (Citigroup) to 4.89% (Wells Fargo); intervals 1.5–2.4 points wide
 - ▶ E.g. Citigroup [3.23, 4.74], Wells Fargo [3.68, 5.71]: the marginal intervals overlap
- Overlap is **not** a test
 - ▶ Estimates from the same days are dependent (here strongly positively correlated)
 - ▶ The interval for the difference can be narrower than either marginal interval
- At $\alpha = 5\%$ the values are smaller (JPMorgan 2.62%) and the ranking changes
 - ▶ Spearman correlation of the 1% and 5% rankings: 0.10; not a robust basis for regulation

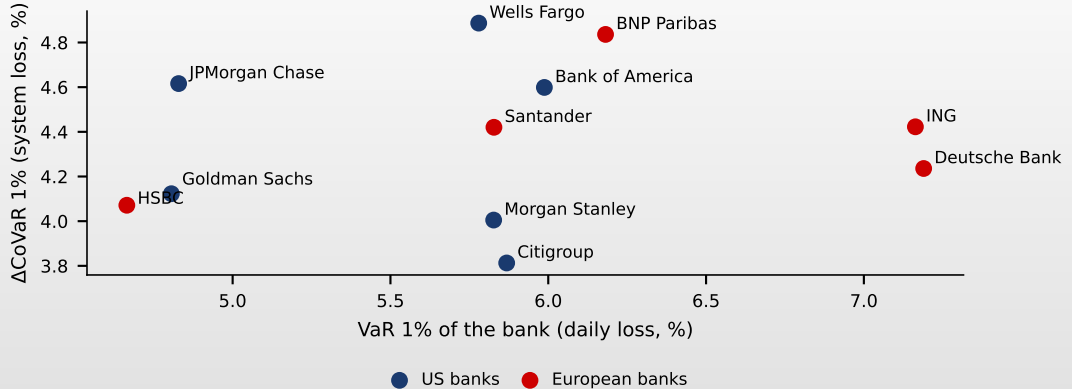
Testing ΔCoVaR 1%

- All tests use the same 999 moving-block bootstrap draws of days for every bank, so estimates of different banks keep their dependence
- (1) No contribution, $H_0: b_\alpha = 0$: Wald test with the sandwich variance of quantile regression [Koenker & Bassett \(1978\)](#)
 - ▶ Kernel (sandwich) t -ratios of $\hat{b}_{1\%}$ from 4.0 (HSBC) to 6.9 (JPMorgan): every bank contributes
- (2) The tail link equals the central link, $H_0: b_\alpha = b_{50\%}$: is the bank special in crises?
 - ▶ Rejected for 2 of 11 banks at 5% (2 after Holm): for most banks the tail link equals the central link
- (3) Equal contributions, $H_0: \Delta\text{CoVaR}_i = \Delta\text{CoVaR}_j$: the same resampled days for **all** banks give the joint distribution
 - ▶ Equal contributions within a region: 4 of 25 pairs differ at 5%, 0 after Holm; joint Wald $p = 0.01$ (US), 0.31 (Europe)
 - ▶ At 5%: 3 of 25 pairs differ, 0 after Holm; joint Wald $p = 0.014$ (US), 0.046 (Europe)
- Full table of slopes and p -values: [Appendix](#); significance and dominance tests for systemic importance: [Castro & Ferrari \(2014\)](#)

Losses at Least Equal to VaR: the Girardi–Ergün CoVaR

- Girardi & Ergün (2013) condition on the bank's loss being **at least** its VaR, not exactly equal to it
 - ▶ $P\left(X^{sys} \leq -\text{CoVaR}_{\alpha}^{\leq} \mid X_i \leq -\text{VaR}_{i,\alpha}\right) = \alpha \iff P\left(X^{sys} \leq -\text{CoVaR}_{\alpha}^{\leq}, X_i \leq -\text{VaR}_{i,\alpha}\right) = \alpha^2$
 - ▶ $\text{CoVaR}_{\alpha}^{\leq}$: CoVaR given a loss at least as large as the VaR; the joint form follows from $P(A \mid B) = P(A, B)/P(B)$ with $P(B) = \alpha$
 - ▶ Benchmark state: a one-standard-deviation event $|X_i - \mu_i| \leq \sigma_i$ (μ_i, σ_i : mean and standard deviation of X_i) instead of the median
- Why it matters
 - ▶ It includes more severe distress than the VaR day itself, and it is monotone in the dependence parameter
 - ▶ It can be **backtested**: the joint hit $\{X^{sys} \leq -\text{CoVaR}_{\alpha}^{\leq}, X_i \leq -\text{VaR}_i\}$ has probability α^2
- Historical estimate at $\alpha = 5\%$: the 5% quantile of the system on the 198 days with $X_i \leq \hat{q}_{5\%}(X_i)$
 - ▶ JPMorgan: $\text{CoVaR}_{5\%}^{\leq} = 7.70\%$ vs the quantile-regression $\text{CoVaR}_{5\%} = 3.81\%$; eleven banks: 7.3–8.2% vs 3.6–4.8%
 - ▶ About twice as large: averaging over the whole tail of the bank is a harsher conditioning event (Seminar 18, A4 derives this under the Normal distribution)

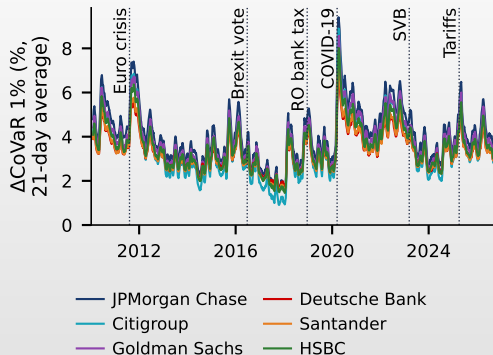
VaR and ΔCoVaR Measure Different Things (1/2)



VaR and ΔCoVaR Measure Different Things (2/2)

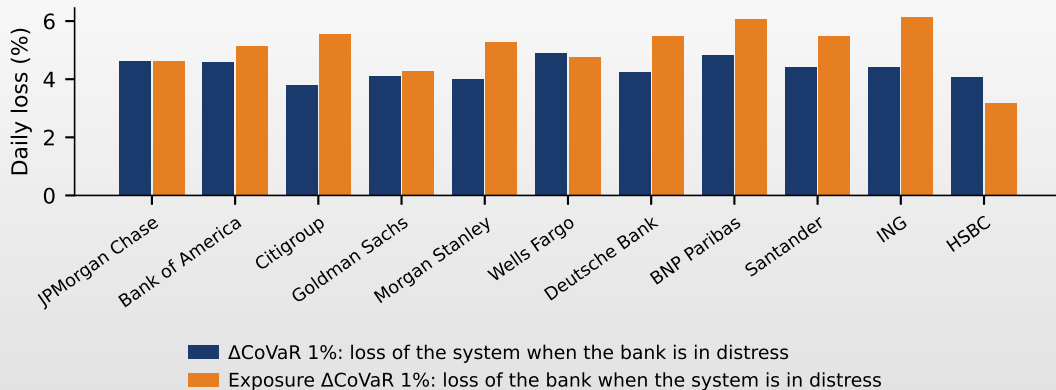
- Horizontal: VaR 1% of the bank; vertical: its ΔCoVaR 1%; eleven banks, 2010–2026
- Spearman rank correlation between VaR and ΔCoVaR : 0.16
 - ▶ Deutsche Bank and ING: the largest VaR (7.19%, 7.16%), only average ΔCoVaR
 - ▶ JPMorgan: one of the smallest VaR (4.83%), one of the largest ΔCoVaR
- Central message of [Adrian & Brunnermeier \(2016\)](#)
 - ▶ Regulating each bank by its own VaR ignores its contribution to the system
- Caveat: with eleven banks and overlapping intervals the correlation is very uncertain (Seminar 18, B4)

ΔCoVaR 1% over Time



- Six banks, 21-day moving averages of the daily ΔCoVaR 1%
 - ▶ State variables M_{t-1} : log VIX, regional index return, system return, all lagged one day [Adrian & Brunnermeier \(2016\)](#)
 - ▶ $\Delta\text{CoVaR}_{\alpha,t} = \hat{b}(\hat{q}_{50\%,t}(X_i) - \hat{q}_{\alpha,t}(X_i))$, with $\hat{q}_{\cdot,t}$ conditional on M_{t-1} ([Appendix](#))
- JPMorgan: 2.16% in 2017, 9.06% in March 2020; peak 10.99% on 2020-03-18
 - ▶ Deutsche Bank: 2.04% in 2017, 6.65% in March 2020
- Low in calm periods, exactly when risk may build up in balance sheets
 - ▶ Hence [Adrian & Brunnermeier \(2016\)](#) propose a forward ΔCoVaR, predicted from leverage, size and maturity mismatch

Contribution versus Exposure (1/2)



Contribution versus Exposure (2/2)

- Two directions of ΔCoVaR 1%
 - ▶ Blue, contribution: loss of the system when the bank is in distress
 - ▶ Orange, exposure: loss of the bank when the system is in distress
- The direction matters: the two numbers differ for most banks
 - ▶ ING: contribution 4.42%, exposure 6.14%; BNP Paribas: 4.84% and 6.07%
 - ▶ HSBC: contribution 4.07%, exposure only 3.17%
- Exposure exceeds contribution for most European banks
 - ▶ Both are associations, not causal effects
- Exposure matters for the bank's own capital; contribution for the regulator

How Much Can We Trust a Ranking?

- Rankings by different systemic measures disagree [Nucera, Schwaab, Koopman & Lucas \(2016\)](#)
 - ▶ Each ranking has wide sampling uncertainty
 - ▶ Combine measures and report rank uncertainty instead of a single ordering
- Model risk: the same bank's measure varies across estimation methods [Danielsson, James, Valenzuela & Zer \(2016\)](#)
 - ▶ In this chapter: LRMES shortcut vs simulation (ratio 0.42–0.90); quantile-regression vs Girardi–Ergün CoVaR (about $\times 2$)
 - ▶ $\alpha = 1\%$ vs 5% : Spearman 0.10
- Effective tail sample αn : 40 days at 1% , 198 at 5% , out of 3 960
 - ▶ Block-bootstrap SE of $\Delta\text{CoVaR } 1\% \approx 0.6 \times$ the kernel SE (median over banks): the two methods disagree
- Out-of-sample backtests of $\text{CoVaR}_{5\%}^{\leq}$ reject for no bank ([Appendix](#))
 - ▶ With about 7.4 expected joint hits their power is low
- What survives: large, tested differences
 - ▶ Regions and the tail slope of some banks, not the order of eleven close point estimates

From Pairs to Networks: the Variance Decomposition (1/2)

- CoVaR looks at one bank and one system; a **network** looks at all pairs at once
 - ▶ Nodes: banks; directed edges: how much of the future uncertainty of bank i comes from shocks to bank j
- Vector autoregression (VAR) of order p for N series y_t : $y_t = c + \sum_{l=1}^p A_l y_{t-l} + \varepsilon_t$, $\text{Cov}(\varepsilon_t) = \Sigma$
 - ▶ Each series is regressed on p lags of all series: c : constants; A_l : $N \times N$ coefficient matrices; ε_t : shocks; Σ : their covariance matrix
- Variance-decomposition networks Diebold & Yilmaz (2009), Diebold & Yilmaz (2012), Diebold & Yilmaz (2014): the approach of this section
- Other networks: Granger causality Billio, Getmansky, Lo & Pelizzon (2012) (▶ [Appendix](#)), tail networks TENET Härdle, Wang & Yu (2016) and the Financial Risk Meter (next section)

From Pairs to Networks: the Variance Decomposition (2/2)

- The H -day forecast error of y_i is a sum of shocks to all series: its variance can be split by source
 - ▶ Moving-average form $y_t = \mu + \sum_{h \geq 0} \Phi_h \varepsilon_{t-h}$: Φ_h is the response of y after h days to a shock today, $\Phi_0 = I$
- Orthogonal (Cholesky) shocks depend on the ordering of the series; the **generalized** decomposition does not [Koop, Pesaran & Potter \(1996\)](#), [Pesaran & Shin \(1998\)](#)
 - ▶ $\theta_{ij}(H) = \frac{\sigma_{jj}^{-1} \sum_{h=0}^{H-1} (e_i^\top \Phi_h \Sigma e_j)^2}{\sum_{h=0}^{H-1} e_i^\top \Phi_h \Sigma \Phi_h^\top e_i}$, normalised: $\tilde{\theta}_{ij} = \theta_{ij} / \sum_j \theta_{ij}$
 - ▶ Numerator: the part of the forecast error variance of i due to a shock in j ; denominator: the total forecast error variance of i
 - ▶ e_i : the i -th unit vector (it picks series i); σ_{jj} : the variance of shock j , the j -th diagonal element of Σ

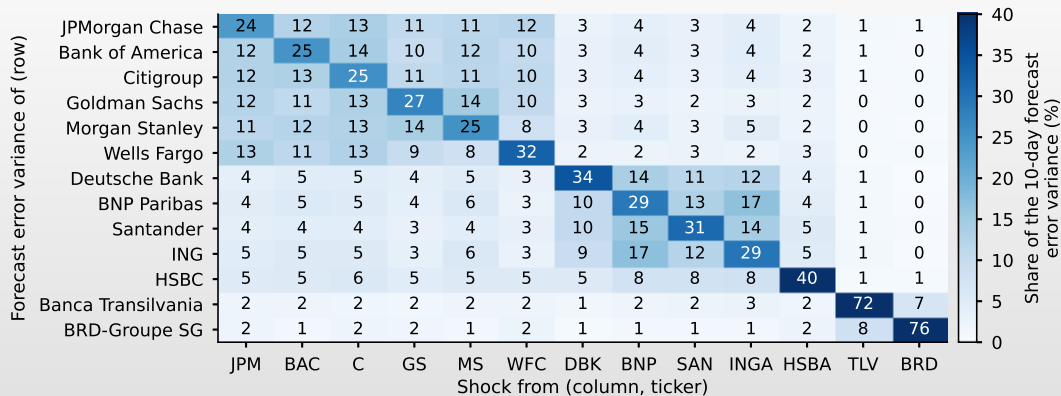
The Connectedness Table

- ▣ $\tilde{\theta}_{ij}$ (in %): the share of the H -day forecast error variance of i due to shocks in j Diebold & Yilmaz (2014)
 - ▶ Each row sums to 100
 - ▶ Before normalisation rows of θ_{ij} do not sum to one: generalized shocks are correlated (Seminar 18, A3)
- ▣ Directional measures
 - ▶ **From** others: $F_i = \sum_{j \neq i} \tilde{\theta}_{ij}$; **to** others: $T_j = \sum_{i \neq j} \tilde{\theta}_{ij}$
 - ▶ **Net**: $T_i - F_i$, positive for a net transmitter
 - ▶ **Pairwise net**: $\tilde{\theta}_{ji} - \tilde{\theta}_{ij} > 0$ means i transmits net to j , drawn as an arrow $i \rightarrow j$
- ▣ **Total connectedness**: $C = \frac{1}{N} \sum_{i \neq j} \tilde{\theta}_{ij}$
 - ▶ The average share of variance that comes from the others, between 0 and 100%

Range Volatility and Our Model

- ▣ **Parkinson volatility** from the daily high H_t and low L_t [Parkinson \(1980\)](#)
 - ▶ $\hat{\sigma}_t^2 = \frac{(\ln H_t - \ln L_t)^2}{4 \ln 2}$, annualised $100\sqrt{252 \hat{\sigma}_t^2}$, used in logs
 - ▶ Unbiased for the variance of a Brownian motion without drift; more efficient than the squared return
- ▣ Our model: VAR on the log volatility of the 13 banks
 - ▶ Order by the Bayesian information criterion ($p = 2$); horizon $H = 10$ days
 - ▶ Time zones: US banks close after European ones; the VAR lags capture next-day transmission

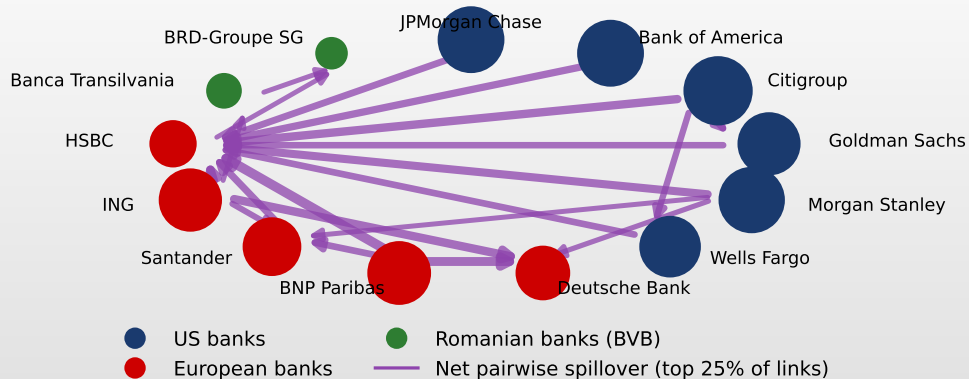
The Connectedness Table of Thirteen Banks (1/2)



The Connectedness Table of Thirteen Banks (2/2)

- Row i , column j : share (%) of the 10-day forecast error variance of bank i due to shocks in bank j
 - ▶ Full sample
- Total connectedness $C = 63.9\%$
 - ▶ On average two thirds of a bank's volatility risk comes from the other banks
 - ▶ On returns instead of volatility: $C = 75.4\%$
- Two regional blocks, own shock included
 - ▶ A US bank receives 84.0% from US shocks, a European bank 72.7% from European shocks
 - ▶ Across the Atlantic: 26.2% from US to a European bank, 15.0% the other way
- Romanian banks: mostly their own shocks
 - ▶ Own share 72.3% (Banca Transilvania), 76.1% (BRD); from US banks 10.3%, from European banks 8.0%

The Net Spillover Network (1/2)

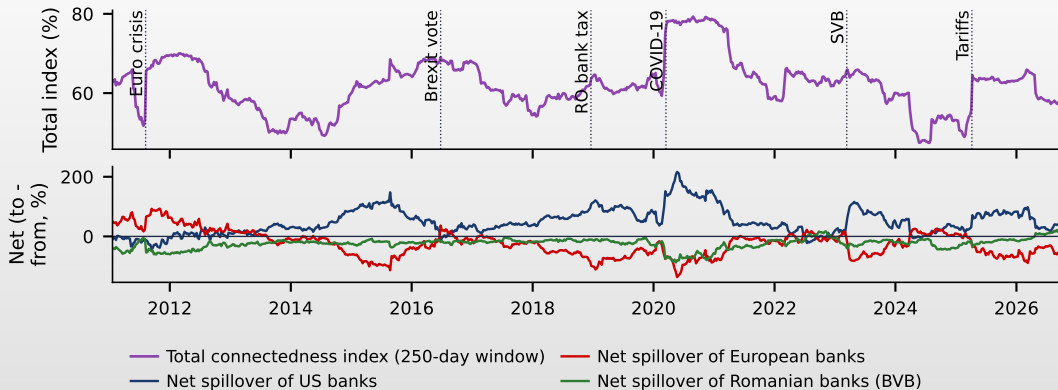


The Net Spillover Network (2/2)

- Arrow $j \rightarrow i$ when j transmits net to i (largest 25% of pairwise net spillovers)
 - ▶ Node size grows with the spillover to others
- Net transmitters: Citigroup (+19.0), Bank of America (+10.9), Morgan Stanley (+9.9), JPMorgan (+9.8)
 - ▶ To others: Citigroup 93.8%, JPMorgan 86.2% (a column sum can exceed 100)
- Net receivers: HSBC (−23.6), BRD (−13.0), Deutsche Bank (−12.6), Banca Transilvania (−12.2)
 - ▶ HSBC receives from almost every bank: it trades in London, between Asia and New York
- Within Europe, BNP Paribas and ING transmit to Deutsche Bank and Santander
- A full-sample network averages quiet and crisis years: next chart over time

 MFM_ch18_spillover

Connectedness over Time (1/2)

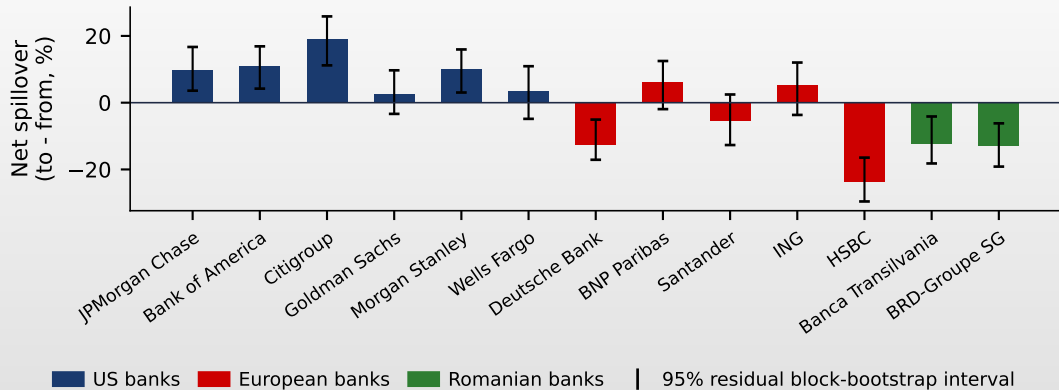


Connectedness over Time (2/2)

- Top: total connectedness on moving 250-day windows ($\text{VAR}(2)$, $H = 10$, every 5 days); bottom: net spillover summed by region
- The total index ranges from 47.4% (2024-07-24) to 79.3% (2020-08-27); mean 62.0%
 - ▶ COVID-19: from 61.8% in 2019 to 78.4% in spring 2020; it jumps within days and decays over a year
 - ▶ After the 2023 bank failures 66.0%; the calm year 2024 53.5%
- US banks are net transmitters in 89% of the windows (average net +48.4)
 - ▶ European banks were net transmitters during the euro crisis of 2011–2012 and mostly net receivers afterwards
- Romanian banks are net receivers on average (−24.1)

 MFM_ch18_spillover_rolling

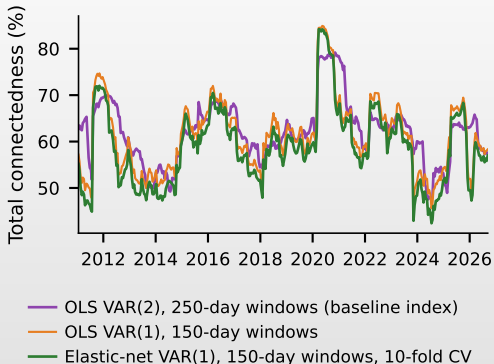
Estimation Error: Bootstrap Intervals for the Network (1/2)



Estimation Error: Bootstrap Intervals for the Network (2/2)

- A VAR(1) on 13 series has 169 slope coefficients
 - ▶ A 250-day window: 250 observations per equation for 13 slopes; the table is a nonlinear function of all of them
- Residual block bootstrap, 999 draws
 - ▶ Resample 20-day blocks of residual vectors, rebuild the series with the estimated VAR, re-estimate, recompute the table
- Full sample, $p = 2$: $C = 63.9\%$, 95% interval [59.9, 65.8]
 - ▶ Peak window (to 2020-08-27): $C = 79.3\%$, [65.9, 83.0]; bootstrap mean 75.8%: small samples also bias C
- 8 of 13 net positions have intervals that exclude zero
 - ▶ For the other 5 the label “transmitter” or “receiver” is not identified
 - ▶ Report intervals; draw only significant pairwise net links

OLS and Elastic-Net Connectedness in Rolling Windows



- Demirer et al. (2018), Eq. (9): each VAR equation by the adaptive elastic net
 - ▶ λ chosen equation by equation by 10-fold cross-validation (definition in the [Appendix](#))
- Full sample: $C = 63.3\%$ (OLS 63.9%); 59% of coefficients non-zero
- Means over 372 windows
 - ▶ Baseline (OLS VAR(2), 250 days) 62.0%; OLS, 150 days 61.1%; elastic net, 150 days 58.7%
- The dynamics agree: correlation 0.82; crisis peaks in the same weeks
 - ▶ Levels depend on window, estimator and lag order; changes over time are robust

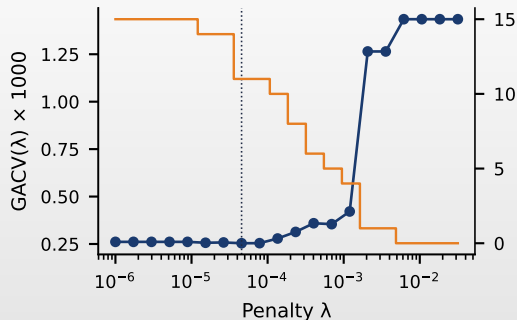
The Financial Risk Meter and the GACV Penalty (1/2)

- For each firm j , a quantile regression at $\tau = 5\%$ of its return on the returns of **all** other firms and on lagged macro variables Mihoci, Althof, Chen & Härdle (2020)
 - ▶ With 100 firms and a 63-day window there are more regressors than days: a penalty is needed
- **LASSO** (Least Absolute Shrinkage and Selection Operator) quantile regression Belloni & Chernozhukov (2011)
 - ▶ $\hat{\beta}_j(\lambda) = \arg \min_{\beta} \frac{1}{n} \sum_t \rho_{\tau} \left(X_{j,t} - \beta_0 - \beta^{\top} Z_{-j,t} \right) + \lambda \|\beta\|_1$
 - ▶ $X_{j,t}$: return of firm j ; $Z_{-j,t}$: returns of the other firms and the lagged macro variables; β_0 : intercept; ρ_{τ} : the check function; $\|\beta\|_1 = \sum_k |\beta_k|$; n : days in the window
 - ▶ λ switches off weak links: a larger λ leaves fewer active coefficients
- **GACV** (Generalized Approximate Cross-Validation) trades fit against the number of active coefficients
 - ▶ $\text{GACV}(\lambda) = \frac{\sum_t \rho_{\tau} \left(X_{j,t} - \hat{X}_{j,t}(\lambda) \right)}{n - \text{df}(\lambda)}$, $\hat{X}_{j,t}(\lambda)$: fitted quantile; $\text{df} =$ number of nonzero coefficients; the selected λ_j minimises GACV

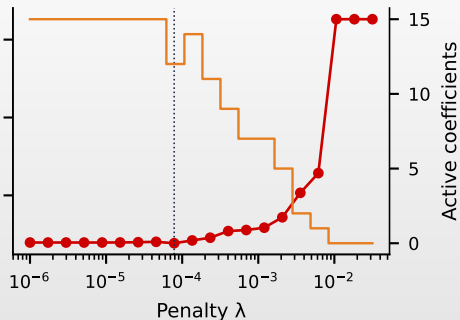
The Financial Risk Meter and the GACV Penalty (2/2)

- The **FRM** (Financial Risk Meter) is the average of the selected λ_j over firms, designed as an indicator of tail interdependence and systemic stress
 - ▶ Built for US financial firms [Mihoci, Althof, Chen & Härdle \(2020\)](#), emerging markets [Ben Amor, Althof & Härdle \(2022\)](#) and China [Wang, Althof & Härdle \(2023\)](#)
- Our version: 13 banks, macro variables S&P 500, Euro Stoxx 50 and VIX at $t - 1$, 63-day windows every 10 days
 - ▶ 15 regressors for 63 days: far fewer than in the published FRM
- Extensions: FRM based on expectiles [Ren, Lu, Li & Härdle \(2022\)](#), for cryptocurrencies [Ren, Althof & Härdle \(2022\)](#) and for the EU energy sector [Toma, Dima, Grecu, Pele & Lessmann \(2026\)](#)

GACV in a Calm and in a Crisis Window (1/2)



—●— Calm window ending 28 Jun 2019
— Active coefficients

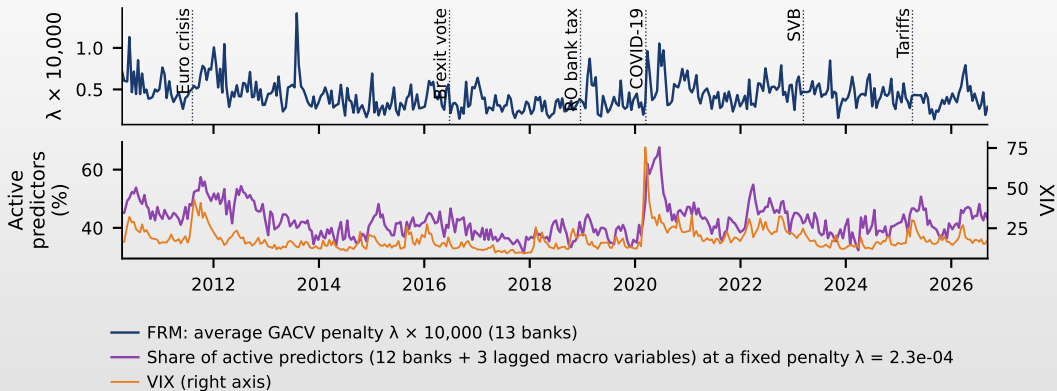


—●— COVID-19 window ending 31 Mar 2020
— Active coefficients

GACV in a Calm and in a Crisis Window (2/2)

- JPMorgan on the other 12 banks and three lagged macro variables, 63 days
 - ▶ Dotted line: selected λ ; orange: active coefficients
- Calm window (to 28 June 2019): $\hat{\lambda} = 4.5 \times 10^{-5}$, 11 of 15 coefficients active
 - ▶ COVID-19 window (to 31 March 2020): $\hat{\lambda} = 7.8 \times 10^{-5}$, 12 active
- In the crisis the whole GACV curve is about three times higher
 - ▶ The check loss is in return units, and crisis returns are larger
- With few regressors the minimum lies at almost no penalty: the selection is noisy

The Financial Risk Meter of Thirteen Banks (1/2)



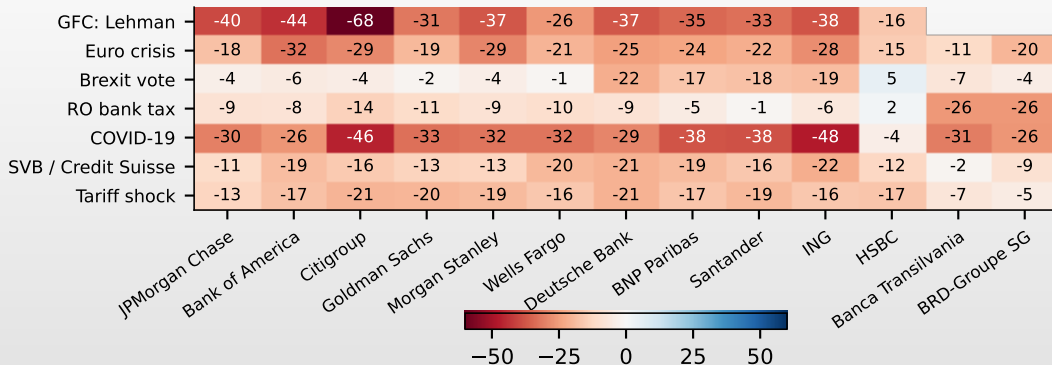
The Financial Risk Meter of Thirteen Banks (2/2)

- Top: average GACV penalty over 13 banks; 390 windows of 63 days
 - ▶ Bottom: share of active predictors at a fixed λ , with the VIX
- The average GACV penalty is noisy: correlation with the VIX only 0.35
 - ▶ 15 regressors and 63 days hardly need a penalty; the published FRM uses about 100 firms
- At a fixed $\lambda = 2.3 \times 10^{-4}$ the share of active predictors follows the VIX more closely (0.62)
 - ▶ From 31% (2017-11-21) to 68% (2020-06-17); average 42%
- Lesson: on this small panel, neither measure is shown to be a reliable systemic-risk indicator
 - ▶ The fixed-penalty count is only an exploratory sensitivity check

Stress Tests and Historical Scenarios

- A stress test: how much would we lose in a **severe but plausible** scenario?
 - ▶ **Historical**: replay a past crisis; **hypothetical**: shock chosen factors
 - ▶ **Reverse**: find the most plausible scenario behind a fatal loss
 - ▶ Supervisory tests: [Federal Reserve stress tests](#) (US), [European Banking Authority](#) (EU); global systemically important banks hold extra capital [Basel Committee \(2018\)](#)
- Seven episodes, each a window of a few weeks around a known shock
 - ▶ Lehman (autumn 2008), euro crisis (summer 2011), Brexit vote (June 2016), Romanian bank tax (December 2018)
 - ▶ COVID-19 (March 2020), SVB and Credit Suisse (March 2023), the tariff shock (April 2025)
- In each window: the **10 consecutive trading days with the lowest cumulative return** of the equal-weighted bank portfolio
 - ▶ We report every bank's return over that interval; ten days match the market-risk horizon ([Chapter 7](#))
 - ▶ 2008 uses the 11 international banks only: the Romanian series start in 2010

Seven Historical Stress Scenarios (1/3)



Return over the worst 10 days of the equal-weighted bank portfolio (%)

Seven Historical Stress Scenarios (2/3)

- Return (%) of each bank over the worst 10 consecutive days of the bank portfolio in each episode; blank: no data

 MFM_ch18_stress

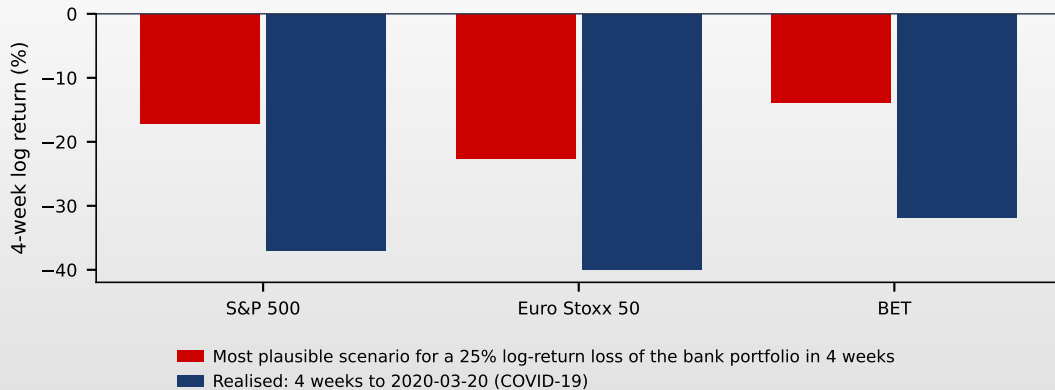
Seven Historical Stress Scenarios (3/3)

- ▣ Portfolio loss over the worst 10 days
 - ▶ Lehman 37.6% (2008-11-10 – 2008-11-21), COVID-19 31.9%, euro crisis 22.3%
 - ▶ Tariffs 16.0%, SVB 14.8%, Romanian bank tax 10.2%, Brexit 7.9%
- ▣ Worst single cells: Citigroup –68% in 2008, ING –48% in March 2020
- ▣ Shocks are local as well as global
 - ▶ Brexit hit mainly European banks
 - ▶ The Romanian bank tax hit Romanian banks hardest (–26%, –26%)
 - ▶ The same December 2018 sell-off cost US banks –8% to –14%
- ▣ A stress test built only on global crises would miss the domestic political shock

Reverse Stress Testing

- Factor model of weekly bank log returns (%): $X_i = \beta_i^\top f + e_i$
 - ▶ Factors $f = (\text{S\&P 500, Euro Stoxx 50, BET})$; β_i : the bank's factor betas; e_i : idiosyncratic return
 - ▶ Over h weeks $f \sim N(0, h\Sigma)$, Σ : weekly factor covariance (no serial covariance assumed)
- Portfolio loss $L = -b^\top f$, b : the average of the β_i
 - ▶ A factor-only approximation: the residual $\sum_i w_i e_i$ is set to zero
- Most plausible scenario with loss ℓ^* : minimise $f^\top (h\Sigma)^{-1} f$ subject to $-b^\top f = \ell^*$
 - ▶ $f^* = -\ell^* \frac{h\Sigma b}{b^\top h\Sigma b}$, $d^* = \frac{\ell^*}{\sqrt{b^\top h\Sigma b}}$
 - ▶ $f^\top (h\Sigma)^{-1} f$: the squared Mahalanobis distance; d^* : portfolio standard deviations; larger means less plausible
- A two-factor example worked step by step: [▶ Appendix](#) and Seminar 18, A7

The Most Plausible Scenario for a 25% Log-Return Loss (1/2)



The Most Plausible Scenario for a 25% Log-Return Loss (2/2)

- Weekly data 2010–2026 (871 weeks), four-week horizon
 - ▶ Red: f^* for $\ell^* = 25\%$; blue: factor moves in the worst four weeks of 2020
- Portfolio betas $b = (0.45, 0.65, 0.18)$; four-week s.d. 5.79%
 - ▶ A 25% log loss (arithmetic 22.1%) needs S&P 500 -17.2% , Euro Stoxx 50 -22.6% , BET -13.9% together
- $d^* = 4.31$: Normal probability 8.0×10^{-6} per four weeks
 - ▶ Yet in the four weeks to 2020-03-20 the portfolio lost 52.0% (model: 48.5%)
- The shape of the scenario is informative; its probability is not
 - ▶ Heavy tails and rising correlations make Normal probabilities far too small

Climate Risk and the ECB Climate Stress Test of 2022



Seat of the European Central Bank, Frankfurt,
2015

Photo: DXR (2015); CC BY-SA 4.0; Wikimedia
Commons

- **Physical risk:** losses from floods, droughts, heat and storms that damage collateral and borrowers
 - ▶ **Transition risk:** losses from the move to a low-carbon economy (carbon prices, regulation, technology)
- “Green swan” risks: potentially extremely disruptive events that could be behind the next systemic financial crisis; backward-looking risk models cannot anticipate them [Bolton et al. \(2020\)](#)
- 104 significant banks took part; a bottom-up projection of losses covered 41 of them [ECB Banking Supervision \(2022\)](#)
- About 60% of banks had no climate risk stress-testing framework; only 20% considered climate risk when granting loans
- Credit and market losses of around EUR 70 billion for the 41 banks, under a short-term disorderly transition and two physical risk scenarios
 - ▶ The ECB states that this significantly understates the risk: data gaps, rudimentary models, no second-round effects

Limits of Stress Tests

- ▣ Scenarios are chosen by people: the crisis that comes is rarely the one that was tested
- ▣ Second-round effects (fire sales, funding runs, counterparty losses) are hard to model and usually left out, as the ECB noted for its climate test
- ▣ Market-based measures (MES, CoVaR, connectedness) react quickly but only after prices move; balance-sheet stress tests look forward but slowly
- ▣ Early-warning research combines both, e.g. entropy-based measures of market risk [Pele, Lazar & Dufour \(2017\)](#)

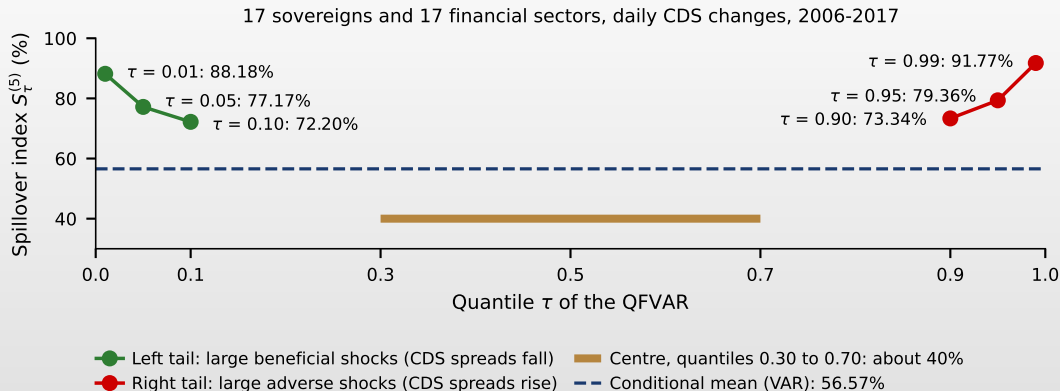
Case Study: Question and Motivation

- Ando, Greenwood-Nimmo & Shin (2022) ask whether the topology of a financial network changes with the **size** of the shocks that hit it (Section 1)
 - ▶ Their version of the question of Diebold & Yilmaz (2014): how much of the future uncertainty of i is due to idiosyncratic shocks from j , as the shock size varies?
- Why it matters: the connectedness of this chapter is estimated at the conditional mean (OLS, LASSO, elastic net Demirer et al. (2018)), i.e. it describes how an average shock spreads
 - ▶ Systemic shocks are larger than average; reading a rise in rolling connectedness as a crisis signal assumes that mean relations also hold in the tails (Section 1)
- Link to this chapter: the quantile regression of CoVaR Adrian & Brunnermeier (2016), Koenker & Bassett (1978), now inside the VAR and its variance decomposition
 - ▶ Giglio, Kelly & Pruitt (2016) showed that systemic risk shifts the lower tail of future output more than its centre; Ando, Greenwood-Nimmo & Shin (2022) ask the same tail-versus-centre question of the network itself
- Application: the credit risk loop between sovereigns and their banks, in which bailouts move bank risk onto the state (Sections 1 and 3)

Case Study: Data and Method

- Data (Section 3): daily first differences of 5-year CDS spreads (basis points) of 17 sovereigns and 17 national financial sectors, $m = 34$, 3 January 2006 – 29 December 2017, 3 129 days
- Quantile factor VAR (QFVAR), Eqs. (5)–(7): $y_t = \mu + \sum_{j=1}^p \Phi_j y_{t-j} + \Lambda f_t + v_t$, each equation fitted at quantile τ by minimising the check loss
 - ▶ y_t : the m CDS changes; μ : constants; Φ_j : VAR coefficient matrices; Λ : factor loadings
 - ▶ Unobserved factors f_t absorb common shocks, so v_t is idiosyncratic; their number by the criterion of [Ando & Bai \(2020\)](#), Eq. (8): 10 at the mean (Figure 1)
- Generalized forecast error variance decomposition (FEVD) of the idiosyncratic shocks at quantile τ , Eq. (17), rows normalised to 1; spillover index $S_\tau^{(h)} = m^{-1} \sum_i F_{i \leftarrow \bullet, \tau}^{(h)}$, Eq. (20)
 - ▶ $F_{i \leftarrow \bullet, \tau}^{(h)}$: the share of the h -day forecast error variance of i that comes from the others, the “from” of the connectedness table; $S_\tau^{(h)}$ is its average over the m series
- Rolling windows of $w = 250$ days (2 880 windows), number of factors fixed at its full-sample value; relative tail dependence $\text{RTD} = S_{0.95}^{(5)} - S_{0.05}^{(5)}$ (Section 4.3)
 - ▶ For CDS changes the right tail ($\tau = 0.95$) holds large adverse shocks, the left tail ($\tau = 0.05$) large beneficial ones
- Full specification (factor criterion, lag order, horizon) and the complete results: ▶ Appendix

Case Study: Spillovers across Quantiles (1/2)

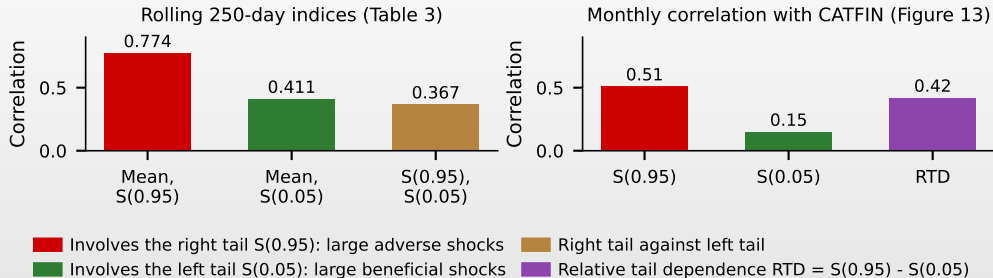


Case Study: Spillovers across Quantiles (2/2)

- Full-sample spillover index $S_{\tau}^{(5)}$ at each quantile τ vs the conditional-mean index
 - ▶ Source: [Ando, Greenwood-Nimmo & Shin \(2022\)](#), Figure 4
- U-shaped profile: about 40% in the centre, 88.18% at $\tau = 0.01$, 91.77% at $\tau = 0.99$
 - ▶ The mean index (56.57%) describes neither regime: too high for small shocks, too low for large ones
- Full sample: almost symmetric tails, $RTD = 79.36 - 77.17 = 2.19$ points
 - ▶ In rolling windows the asymmetry is stronger and varies (next chart)
- Consequence for stress testing
 - ▶ Mean connectedness understates how much of a sector's 5-day forecast error variance comes from other sectors in a crisis

 [MFM_ch18_quantile_connectedness](#)

Case Study: The Tails Move Apart (1/2)



Left: correlations between the rolling indices ($w = 250$ days, $h = 5$); right: monthly correlations with CATFIN; source: Ando, Greenwood-Nimmo & Shin (2022), Table 3 and Figure 13

CATFIN: the average of three VaR 1% estimates for financial stocks Allen, Bali & Tang (2012)

Case Study: The Tails Move Apart (2/2)

- The mean index follows the adverse tail (0.774 with $S_{0.95}$) far more than the beneficial tail (0.411 with $S_{0.05}$)
 - ▶ A rise in mean connectedness is mainly a rise in bad-news spillovers
- The two tails share little variation (0.367)
 - ▶ Two indicators, not one index measured twice
 - ▶ Hence RTD is informative: about ± 15 points in rolling windows, against 2.19 on the full sample
- Against CATFIN: 0.51 for $S_{0.95}$, 0.42 for RTD, only 0.15 for $S_{0.05}$
 - ▶ CATFIN moves with adverse-shock spillovers more than with beneficial ones

Case Study: Replicate, Limits, Extend

- Replicate with the course data: daily closes of the 13 banks of this chapter (course dataset; US, euro area and United Kingdom, Romania), 2010–2026; y_t = daily loss $-r_t$ in %, so that the right tail is the adverse one, as for CDS
 - ▶ Data limitation to state: equity losses instead of CDS spreads, 13 banks and no sovereigns ($m = 13$ instead of 34), different closing times of US, European and Romanian markets
- Limits stated by the authors: the lag order of the mean is imposed at every quantile, and the quantile is held fixed along the horizon (Sections 2.1–2.2, footnote 6)
- Open extension
 - ▶ Does the RTD of the 13 banks rise before March 2020 and the March 2023 bank failures?
 - ▶ Does it predict bank stress beyond current volatility and the mean-based index of this chapter?
 - ▶ For forecasting, select lag order and factors on a training sample only

An Open Question

Does network connectedness measure systemic risk, or only volatility?

- Connectedness indices are monitored as early-warning tools
 - ▶ An index from a VAR on volatilities may co-move with volatility without adding predictive information
 - ▶ Live research: regularised VAR networks [Demirer et al. \(2018\)](#); tail connectedness differs from mean connectedness [Ando, Greenwood-Nimmo & Shin \(2022\)](#)
- Formal version: $y_{t+h} = a + \beta_C C_t + \gamma \ln \sigma_t + e_{t+h}$, $H_0: \beta_C = 0$, $H_1: \beta_C > 0$
 - ▶ y_{t+h} : future bank stress; C_t : total connectedness; σ_t : current volatility
 - ▶ Evidence against: the CI for β_C rules out economically relevant effects; non-significance alone is not enough
- Identification: an edge is a share of forecast error variance, not a causal exposure
 - ▶ A common shock raises volatility and connectedness together, with no bank infecting another

The Discovery Loop, Step by Step

- **Literature:** map what is known, then verify every reference
 - ▶ Resolve each DOI on [Crossref](#) and check that the paper really controls for volatility; search with Semantic Scholar, Elicit or Google Scholar
- **Hypotheses:** competing readings of a rise in connectedness, each with its own test
 - ▶ E.g. a common shock fades with volatility; contagion has a direction: shocks of net transmitters precede those of receivers
- **Data and code:** on the course data, replicating a known number first
 - ▶ Replicate first: mean 62.0% and maximum 79.3% (2020-08-27), as in this chapter
- **Robustness,** fixed before the predictive regressions are run
 - ▶ Every cell of the grid is reported; the family is corrected with Holm or controlled with the FDR
- **Critique:** ask the assistant to attack the result
 - ▶ Expect: add current volatility as a control, then test the increment of R^2 and of the out-of-sample loss
- **Report:** standardised effects with HAC t -statistics, every cell of the grid, what failed
 - ▶ The prompts that shaped the design go into AI_USE.md
- Reusable prompts for every step and the researcher's checklist: ▶ Appendix

Mini-Case: Is Connectedness More Than Volatility?

- Typical assistant proposal: “high total connectedness warns of future bank stress”
 - ▶ Tested on 731 weekly dates, 2011–2026
 - ▶ y_{t+h} : log mean Parkinson volatility of the 13 banks over the next $h = 63$ days; control: the same over the last 20 days
 - ▶ Standardised variables; Newey–West with 13 lags
- Connectedness alone: $\hat{\beta}_C = 0.40$ ($t = 4.63$), $R^2 = 0.16$: the proposal looks right
- With current volatility: $\hat{\beta}_C = 0.08$ ($t = 0.75$), $\hat{\gamma} = 0.53$ ($t = 4.13$)
 - ▶ $R^2 = 0.343$ vs 0.339 with volatility alone; correlation of C_t with volatility 0.59
 - ▶ Next 63-day maximum drawdown: $\hat{\beta}_C = -0.10$ ($t = -1.32$) even as the only regressor
- Interpretation: given current volatility, connectedness adds no significant predictive content
 - ▶ Not proof of zero: in our sample the persistence of volatility explains most of the predictive content
 - ▶ Network information may still matter in the tails [Ando, Greenwood-Nimmo & Shin \(2022\)](#) or for the direction of transmission

Your Turn: a Project Seed

□ Question

- ▶ Does **tail** connectedness predict bank distress beyond volatility, out of sample?

□ Replicate first

- ▶ Our rolling index: mean 62.0%, spring 2020 78.4%
- ▶ One headline connectedness figure of [Demirer et al. \(2018\)](#) on their sample

□ Extend

- ▶ Quantile connectedness at $\tau = 5\%$ [Ando, Greenwood-Nimmo & Shin \(2022\)](#); out-of-sample evaluation from 2018; MES and SRISK as alternative targets

□ Deliverables

- ▶ A grid of specifications fixed in writing before the regressions; a notebook that regenerates every number
- ▶ HAC inference and out-of-sample losses for every cell

□ AI log ([templates](#))

- ▶ AI_USE.md: the prompts that shaped the design
- ▶ AI_ERRORS.md: each wrong claim (a VAR on price levels, OLS t -statistics on overlapping windows) and the test that caught it

Which Bank Is Most Systemic? (1/2)

Bank	VaR 1%	MES 5%	$\Delta\text{CoVaR 1\%}$	Net spillover	L^*
JPMorgan	4.83	3.22	4.62	9.8	7.0
Bank of America	5.99	3.85	4.60	10.9	6.2
Citigroup	5.87	4.03	3.81	19.0	6.0
Goldman Sachs	4.81	3.34	4.12	2.4	6.8
Morgan Stanley	5.83	4.04	4.00	9.9	6.0
Wells Fargo	5.78	3.32	4.89	3.6	6.8
Deutsche Bank	7.19	4.54	4.24	-12.6	6.1
BNP Paribas	6.18	4.53	4.84	6.0	6.1
Santander	5.83	4.23	4.42	-5.4	6.4
ING	7.16	4.81	4.42	5.2	5.8
HSBC	4.66	2.54	4.07	-23.6	8.3
Banca Transilvania	5.06	3.14	-	-12.2	6.5
BRD-Groupe SG	4.42	2.69	-	-13.0	7.1

□ Daily losses in %; net spillover of volatility (to minus from); L^* : break-even leverage with $k = 8\%$

Which Bank Is Most Systemic? (2/2)

- Rank correlations across the eleven international banks
 - ▶ MES and ΔCoVaR : -0.01 ; MES and net spillover: 0.04
 - ▶ ΔCoVaR and net spillover: -0.01 ; VaR and MES: 0.88
- No single ranking
 - ▶ Supervisors combine several indicators with size and balance-sheet data, as in the [Basel Committee \(2018\)](#) methodology
- Romanian banks
 - ▶ Moderate MES, net receivers of volatility
 - ▶ Break-even leverage close to that of US banks

Key Takeaways

- Systemic risk lives in the joint tail
 - ▶ Exposure (MES, SRISK), contribution (ΔCoVaR), connectedness (spillover and tail networks)
- European banks: the largest MES in our sample
 - ▶ US banks are the main net transmitters of volatility
- VaR and ΔCoVaR rank banks differently
 - ▶ Rankings are uncertain: test differences, report intervals, state αn
- Market-based measures jump in crises
 - ▶ Stress tests and balance sheets are needed to see risk building up
- Romanian banks: mostly their own shocks
 - ▶ They receive more volatility spillover than they transmit

Common Pitfalls

- ▣ Reading ΔCoVaR as causal: a common shock creates the same pattern
- ▣ Reading overlapping intervals as “equal” and separate ones as “different”
 - ▶ Test the difference on joint draws and correct for the number of pairs
- ▣ Including the bank itself in the system portfolio
 - ▶ Its own return adds a mechanical self-contribution to the link
- ▣ Orthogonal variance decompositions whose results depend on the order of the banks
- ▣ Reporting SRISK without balance sheets, or the $1 - e^{-18\text{MES}}$ shortcut as the published measure
- ▣ Drawing every Granger edge significant at 5% out of 156 tests, or net labels without intervals

Appendix: MES as an Euler Contribution: Proof

- Portfolio $X_p = \sum_i w_i X_i$ with a continuous distribution; $ES_\alpha(X_p) = -\mathbb{E}[X_p \mid X_p \leq q_\alpha(X_p)]$
 - ▶ ES_α is positively homogeneous of degree 1 in w , so by Euler's theorem $ES_\alpha(X_p) = \sum_i w_i \partial ES_\alpha / \partial w_i$
 - ▶ Differentiating $-\frac{1}{\alpha} \mathbb{E}[X_p 1\{X_p \leq q_\alpha(w)\}]$: the term in $\partial q_\alpha / \partial w_i$ is $-\frac{1}{\alpha} q_\alpha f_p(q_\alpha) \partial q_\alpha / \partial w_i + \frac{1}{\alpha} q_\alpha f_p(q_\alpha) \partial q_\alpha / \partial w_i = 0$
 - ▶ Hence $\partial ES_\alpha / \partial w_i = -\mathbb{E}[X_i \mid X_p \leq q_\alpha(X_p)] = MES_{i,\alpha}$ with $m = p$ [Gouriéroux, Laurent & Scaillet \(2000\)](#), [Scaillet \(2004\)](#)
- Check on our data: equal-weighted portfolio of the six US banks, $\alpha = 5\%$ (198 tail days)
 - ▶ $ES_{5\%}$ of the portfolio = 4.32% = average of the six MES conditional on the portfolio (4.32%): exact, also for the historical estimator
 - ▶ Conditioning on the S&P 500 instead gives an average MES of 3.63% (JPMorgan: 3.22% vs 3.85%): the identity fails when $m \neq p$
- Under the bivariate Normal distribution with zero means, $MES_{i,\alpha} = \beta_i ES_\alpha(X_m)$: a beta times a constant (Seminar 18, A2)

Appendix: LRMES Simulation: Innovations and Recursions

- Standardised residuals $\epsilon_{it} = r_{it}/\sigma_{it}$, $\epsilon_{mt} = r_{mt}/\sigma_{mt}$; S is their sample correlation matrix, the long-run target of Q_t
 - ▶ Bank innovation orthogonal to the market: $\xi_{it} = (\epsilon_{it} - \rho_t \epsilon_{mt})/\sqrt{1 - \rho_t^2}$ [Brownlees & Engle \(2017\)](#), Appendix A
- Each simulated day s : draw a past date τ , take the pair $(\xi_{i\tau}, \epsilon_{m\tau})$ and rebuild the bank shock with the **simulated** correlation
 - ▶ $\epsilon_{i,t+s} = \rho_{t+s} \epsilon_{m\tau} + \sqrt{1 - \rho_{t+s}^2} \xi_{i\tau}$, then update σ^2 and Q with the new returns
- Restrictions: $\omega > 0$, $\alpha \geq 0$, $\alpha + \gamma \geq 0$, $\beta \geq 0$, $\alpha + \gamma/2 + \beta < 1$ (finite variance for symmetric innovations; in general $\alpha + \gamma \mathbb{E}[\epsilon^2 \mathbf{1}\{\epsilon < 0\}] + \beta < 1$); $a, b > 0$, $a + b < 1$, so that $\sigma_t^2 > 0$ and Q_t is positive definite
 - ▶ Initialisation: $Q_1 = S$ in estimation; the simulation starts from σ_{T+1}^2 and Q_{T+1} filtered to the last date

Appendix: Quantile Regression: Asymptotics

- Asymptotics for a fixed α , with $\hat{\theta}_\alpha = (\hat{a}_\alpha, \hat{b}_\alpha)^\top$, n days, $x_t = (1, X_{i,t})^\top$ and f_t the conditional density of Y at its α -quantile
 - ▶ $\sqrt{n}(\hat{\theta}_\alpha - \theta_\alpha) \Rightarrow N\left(0, \alpha(1 - \alpha) D_1^{-1} D_0 D_1^{-1}\right)$, $D_0 = \mathbb{E}[x_t x_t^\top]$, $D_1 = \mathbb{E}[f_t x_t x_t^\top]$
 - ▶ Assumes independent observations (or scores $x_t(\alpha - 1\{u_t < 0\})$ that are a martingale difference), $f_t > 0$ near the quantile and D_0 , D_1 positive definite; with serially dependent scores D_0 needs a long-run (HAC) estimate or a block bootstrap
 - ▶ The **sparsity** $1/f_t$ must be estimated (kernel or bandwidth); at $\alpha = 1\%$ few observations lie near the quantile, so this estimate is noisy
- The slope b_α can change with α : the bank may matter more for the system in the tail than in the centre; least squares gives only the mean slope

Appendix: Extremal Quantiles: Why $\alpha = 1\%$ Is Hard

- The effective tail sample is αn : 40 days at 1% and 198 at 5% out of 3 960
 - ▶ The Normal approximation above assumes $\alpha n \rightarrow \infty$ fast; with αn of a few dozen it can be poor, and extremal-quantile theory (extreme-value limits, subsampling) is the alternative built for this case [Chernozhukov & Fernández-Val \(2011\)](#)
- In our data the block-bootstrap standard error of $\Delta\text{CoVaR } 1\%$ is about 0.6 times the kernel-based standard error (median over banks): the two methods give clearly different uncertainty
 - ▶ The kernel-based standard error rests on a sparsity estimate from the few observations near the 1% quantile; the bootstrap redraws the same few tail days: neither is reliable when αn is this small
- Practical rule: report $\alpha = 5\%$ and $\alpha = 1\%$ with αn ; expect the 1% ranking to be the noisier one
 - ▶ The low Spearman correlation between the 1% and 5% rankings (0.10) may reflect estimation noise, genuine differences between the two tail levels, or both

Appendix: Testing ΔCoVaR : Three Hypotheses

- (1) No contribution, $H_0: b_\alpha = 0$: Wald test with the sandwich variance of quantile regression [Koenker & Bassett \(1978\)](#)
 - ▶ Tests of linear hypotheses across quantiles use the joint asymptotics of $(\hat{b}_\alpha, \hat{b}_{50\%})$ [Koenker & Bassett \(1982\)](#)
- (2) The tail link equals the central link, $H_0: b_\alpha = b_{50\%}$: is the bank special in crises?
 - ▶ We estimate the variance of $\hat{b}_\alpha - \hat{b}_{50\%}$ from 999 moving-block bootstrap draws that keep the pairs of days together
- (3) Equal contributions, $H_0: \Delta\text{CoVaR}_i = \Delta\text{CoVaR}_j$: the same resampled days for **all** banks give the joint distribution
 - ▶ $z_{ij} = (\widehat{\Delta\text{CoVaR}}_i - \widehat{\Delta\text{CoVaR}}_j) / \widehat{\text{se}}^*(\cdot)$ for each pair; Holm correction over the pairs [Holm \(1979\)](#); a joint Wald test that all are equal
 - ▶ Significance and dominance tests for systemic importance: [Castro & Ferrari \(2014\)](#)

Appendix: Testing ΔCoVaR 1%: Results

Bank	ΔCoVaR	$\hat{b}_{1\%}$	z	$\hat{b}_{50\%}$	$\hat{b}_{1\%} - \hat{b}_{50\%}$	p
JPMorgan	4.62	0.94	6.9	0.96	-0.02	0.66
Bank of America	4.60	0.76	6.5	0.78	-0.02	0.73
Citigroup	3.81	0.64	4.7	0.76	-0.11	0.00
Goldman Sachs	4.12	0.85	5.6	0.84	0.00	0.99
Morgan Stanley	4.00	0.68	4.5	0.73	-0.05	0.28
Wells Fargo	4.89	0.84	4.3	0.78	0.06	0.17
Deutsche Bank	4.24	0.59	4.1	0.60	-0.02	0.67
BNP Paribas	4.84	0.77	6.5	0.74	0.03	0.37
Santander	4.42	0.75	4.5	0.69	0.07	0.07
ING	4.42	0.61	4.4	0.71	-0.10	0.00
HSBC	4.07	0.87	4.0	0.86	0.01	0.96

□ z : kernel (sandwich) t -ratio of $\hat{b}_{1\%}$, valid only for serially independent scores: every bank contributes; p : bootstrap test of $b_{1\%} = b_{50\%}$, significant for 2 of 11 banks (2 after Holm)

- ▶ Equal contributions within a region: 4 of 25 pairs differ at 5%, 0 after Holm; joint Wald $p = 0.01$ (US), 0.31 (Europe)
- ▶ At 5%: 3 of 25 pairs differ, 0 after Holm; joint Wald $p = 0.014$ (US), 0.046 (Europe)

Appendix: Time-Varying ΔCoVaR with State Variables

- State variables M_{t-1} : log VIX, the return of the regional index and of the system, all lagged one day [Adrian & Brunnermeier \(2016\)](#)
 - ▶ $q_\alpha(X_{i,t} \mid M_{t-1}) = a_i + c_i^\top M_{t-1}$ at α and at 50% (a_i, c_i : intercept and coefficients of the bank's quantile regression)
 - ▶ $q_\alpha(X_t^{\text{sys}} \mid X_{i,t}, M_{t-1}) = a + b X_{i,t} + c^\top M_{t-1}$
- Then $\Delta\text{CoVaR}_{\alpha,t} = \hat{b} (\hat{q}_{50\%,t}(X_i) - \hat{q}_{\alpha,t}(X_i))$
 - ▶ It moves with the bank's conditional VaR, which moves with the VIX
- The slope $\hat{b}$ is constant: time variation comes from the state, not from the link

Appendix: Backtesting CoVaR and MES

- Systemic measures are conditional risk measures, so they need **conditional** backtests [Banulescu-Radu, Hurlin, Leymarie & Scaillet \(2021\)](#)
 - ▶ For $\text{CoVaR}^{\leq}$: hit $I_t = 1\{X_t^{\text{sys}} \leq -\text{CoVaR}_t^{\leq}, X_{i,t} \leq -\text{VaR}_{i,t}\}$ with $P(I_t = 1) = \alpha^2$ under H_0 if both the bank VaR and the conditional quantile are calibrated; Kupiec and Christoffersen tests on I_t ([Chapter 8](#)), plus a separate check of the bank VaR rate
 - ▶ For MES: the cumulative joint violation process of [Banulescu-Radu, Hurlin, Leymarie & Scaillet \(2021\)](#)
- Our out-of-sample check: historical $\text{CoVaR}_{5\%}^{\leq}$ on an expanding window from day 1 000, 2 960 forecast days per bank
 - ▶ Expected joint hits 7.4; observed 5–9 across the eleven banks (JPMorgan 9); smallest Kupiec $p = 0.35$: no rejection
 - ▶ Power is limited: if the true joint rate were twice α^2 , the 5% Kupiec test would reject in 62% of samples of this length with independent hits, so even a doubling is missed about one time in three

Appendix: VAR Assumptions and the Lag Order of the Rolling Index

- Vector autoregression (VAR) of order p for N series y_t : $y_t = c + \sum_{l=1}^p A_l y_{t-l} + \varepsilon_t$, $\text{Cov}(\varepsilon_t) = \Sigma$
 - ▶ Assumptions: stable VAR (roots of $\det(I - \sum_l A_l z^l)$ outside the unit circle), ε_t serially uncorrelated with finite, positive definite Σ
 - ▶ Moving-average form: $y_t = \mu + \sum_{h \geq 0} \Phi_h \varepsilon_{t-h}$, $\Phi_0 = I$, $\Phi_h = \sum_{l=1}^p A_l \Phi_{h-l}$ ($\Phi_h = 0$ for $h < 0$); e_i : the i -th unit vector
- Lag order: $p = 2$ is held fixed in every window so that the index does not jump when the selected order changes; BIC picks $p = 1$ in 100% of the windows, and with $p = 1$ the index has correlation 0.99 with the one shown

Appendix: Regularised VARs: the Elastic Net

- [Demirer et al. \(2018\)](#), Eq. (9): each VAR equation by the adaptive elastic net, λ chosen equation by equation by 10-fold cross-validation
 - ▶ $\hat{\beta} = \arg \min_{\beta} \sum_t (y_t - \sum_i \beta_i x_{it})^2 + \lambda \sum_i w_i (\frac{1}{2} |\beta_i| + \frac{1}{2} \beta_i^2)$, $w_i = 1/|\hat{\beta}_{i,OLS}|$
 - ▶ x_{it} : the lagged series (regressors); λ : penalty strength; w_i : adaptive weights, larger for coefficients small in OLS; the $|\beta_i|$ part sets coefficients to zero, the β_i^2 part shrinks them
 - ▶ Rolling windows of 150 days with the plain elastic net, $w_i = 1$ (Section 5, footnote 15); forecast horizon $H = 10$
- Why: selection and shrinkage in the VAR coefficients, without forcing a sparse **network** (links also come through Σ)
 - ▶ The paper does not state the lag order in the text; we keep the BIC order of our OLS VAR, $p = 2$
- Full sample, adaptive elastic net: $C = 63.3\%$ (OLS 63.9%); 59% of the coefficients are non-zero
 - ▶ With about 4 000 days the penalty hardly matters; it matters in short windows (rolling-window chart in the main text)

Appendix: Granger-Causality Networks and Multiple Testing

- Billio, Getmansky, Lo & Pelizzon (2012) draw an edge $j \rightarrow i$ when j Granger-causes i at 5%; with 13 banks this is $13 \times 12 = 156$ tests
 - ▶ Even with no links at all, about 7.8 edges would appear by chance
- Our tests: bivariate regressions, one lag, heteroskedasticity-robust Wald statistics
 - ▶ Daily returns: 57 edges at 5%, 33 after Holm [Holm \(1979\)](#), 55 with the false discovery rate controlled at 5% [Benjamini & Hochberg \(1995\)](#)
 - ▶ Log volatility, highly persistent: one extra lag that is not tested [Toda & Yamamoto \(1995\)](#), as in the lag-augmented tests of [Shi, Hurn & Phillips \(2020\)](#): 153, 150 and 153 edges
- Holm controls the chance of any false edge under arbitrary dependence; the FDR is the **expected** share of false edges among those drawn, and BH guarantees it for independent tests [Benjamini & Hochberg \(1995\)](#), while tests on overlapping bank pairs are dependent
- Daily returns of banks in different time zones: many “causal” edges only reflect that New York closes after Frankfurt

Appendix: Worked Example: Reverse Stress Test with Two Factors

- Betas $b = (1.2, 0.8)$; four-week covariance of the factors $h\Sigma = \begin{pmatrix} 16 & 6 \\ 6 & 9 \end{pmatrix} (\%^2)$; target loss $\ell^* = 20\%$
 - ▶ $h\Sigma b = (1.2 \cdot 16 + 0.8 \cdot 6, 1.2 \cdot 6 + 0.8 \cdot 9) = (24, 14.4)$
 - ▶ $b^\top h\Sigma b = 1.2 \cdot 24 + 0.8 \cdot 14.4 = 40.32$, standard deviation 6.35%
- Most plausible scenario and its distance
 - ▶ $f^* = -20 \cdot (24, 14.4)/40.32 = (-11.9\%, -7.1\%)$, $d^* = 20/6.35 = 3.15$
 - ▶ Normal probability of a loss at least this large: $\Phi(-3.15) = 0.08\%$ per four weeks
- The scenario loads more on the factor with the larger beta and variance, and both factors fall together because they are correlated

Appendix: Case Study: Data and Method

- Data (Section 3): daily first differences of 5-year CDS spreads (basis points) of 17 sovereigns and 17 national financial sectors, $m = 34$, 3 January 2006 – 29 December 2017, 3 129 days
 - ▶ A financial sector = equal-weighted CDS of its banks and insurers with assets of at least USD 10 billion, 199 institutions in all
- Quantile factor VAR (QFVAR), Eqs. (5)–(7): $y_t = \mu + \sum_{j=1}^p \Phi_j y_{t-j} + \Lambda f_t + v_t$, each equation fitted at quantile τ by minimising the check loss
 - ▶ Unobserved factors f_t absorb common shocks, so v_t is idiosyncratic; their number by the criterion of [Ando & Bai \(2020\)](#), Eq. (8): 10 at the mean (Figure 1)
 - ▶ Residual pairs with $|\rho| < 0.25$: 91.6% with factors, fewer than half without (Figure 2)
- Generalized FEVD of the idiosyncratic shocks at quantile τ , Eq. (17), rows normalised to 1; spillover index $S_\tau^{(h)} = m^{-1} \sum_i F_{i \leftarrow \bullet, \tau}^{(h)}$, Eq. (20)
 - ▶ VAR(1) chosen by BIC at the mean and kept at every quantile; $h = 5$ days; the quantile is held fixed along the forecast horizon (Sections 2.2 and 4)
- Rolling windows of $w = 250$ days (2 880 windows), number of factors fixed at its full-sample value; relative tail dependence $\text{RTD} = S_{0.95}^{(5)} - S_{0.05}^{(5)}$ (Section 4.3)
 - ▶ For CDS changes the right tail ($\tau = 0.95$) holds large adverse shocks, the left tail ($\tau = 0.05$) large beneficial ones

Appendix: Case Study: Key Results

- Full-sample spillover index (Figure 4): about 40% over the central 40% of the distribution; 72.20%, 77.17%, 88.18% at $\tau = 0.10, 0.05, 0.01$; 73.34%, 79.36%, 91.77% at $\tau = 0.90, 0.95, 0.99$; 56.57% at the mean
 - ▶ Small shocks stay local; large shocks, good or bad, spread; the mean averages the two regimes into one number
- Of the 1122 bilateral spillovers, 568 are below 1% at the mean and 829 at the median (Figure 7)
 - ▶ Away from the mean, the network has many weak links rather than a few strong ones; in the tails these weak links add up to strong average spillovers
- Rolling windows (Table 3): the mean index correlates 0.774 with $S_{0.95}$ but 0.411 with $S_{0.05}$; the two tails correlate only 0.367
 - ▶ The tails move apart: RTD is 2.19 percentage points on the full sample but reaches about ± 15 percentage points in rolling windows (Section 4.3, Figure 11)
- Monthly correlation with CATFIN [Allen, Bali & Tang \(2012\)](#), the average of three VaR 1% estimates for financial stocks (Figure 13): 0.51 for $S_{0.95}$, 0.15 for $S_{0.05}$, 0.42 for RTD
 - ▶ RTD rises with the Bear Stearns fund losses of 2007 and falls after the US bank rescue programme and the negative rates of the ECB: a fragility signal the mean hides

Appendix: Case Study: Replication Specification and Limits Stated by the Authors

- Replicate with the course data: daily closes of the 13 banks of this chapter (course dataset; US, euro area and United Kingdom, Romania), 2010–2026; y_t = daily loss $-r_t$ in %, so that the right tail is the adverse one, as for CDS
 - ▶ Specification: VAR order by BIC, factors by Eq. (8), $h = 5$, $\tau \in \{0.05, 0.5, 0.95\}$ and the mean, $w = 250$, RTD
 - ▶ Data limitation to state: equity losses instead of CDS spreads, 13 banks and no sovereigns ($m = 13$ instead of 34), different closing times of US, European and Romanian markets
- Limits stated by the authors: the lag order of the mean is imposed at every quantile, and the quantile is held fixed along the horizon (Sections 2.1–2.2, footnote 6)
 - ▶ 250 days are too few to select factors window by window, so their number is fixed (Section 4.3, Appendix B); shorter windows leave too few effective tail observations
 - ▶ Only direct spillovers of idiosyncratic shocks are measured, not propagation through the common factors (footnote 2); Greece is left out for a gap in its CDS data (Section 3, Appendix A)

Appendix: The Discovery Loop, Reusable Prompts

- **Literature:** map what is known, then verify every reference
 - ▶ “Which papers test whether connectedness indices predict bank distress out of sample, beyond volatility? Give DOI, sample, horizon and controls.”
- **Hypotheses:** competing readings of a rise in connectedness, each with its own test
 - ▶ “Give competing interpretations of a rise in total connectedness (common shock, contagion, artefact of rolling windows) and a test that discriminates between them.”
- **Data and code:** on the course data, replicating a known number first
 - ▶ “Fit a VAR(2) to the log Parkinson volatilities of the 13 banks of the course data on 250-day windows and compute the generalized total connectedness with $H = 10$.”
- **Robustness,** fixed before the predictive regressions are run
 - ▶ “Pre-register: horizons 21/63/126 days, volatility and VIX controls, windows 150/250/500 days, HAC lags at least the overlap, out-of-sample evaluation from 2018.”
- **Critique:** ask the assistant to attack the result
 - ▶ “Act as a referee: is the predictive power of connectedness just the persistence of volatility? Which regression would show it?”

Appendix: What the Researcher Must Check

- **Non-stationary inputs:** a VAR on price levels, or on trending series, produces spurious connectedness
 - ▶ Test: unit-root tests on every input; all eigenvalues of the companion matrix of the fitted VAR strictly inside the unit circle in every window
- **Ordering:** Cholesky decompositions change with the order of the banks
 - ▶ Test: permute the columns; the generalized table must not change
- **Look-ahead:** a window labelled by its first date, or a lag order chosen on the full sample
 - ▶ Test: the index at t must not move when data after t are deleted
- **Overlap:** rolling indices and forward outcomes overlap, so OLS t -statistics are inflated
 - ▶ Test: Newey–West lags at least equal to the overlap, or a block bootstrap
- **Dimension:** 13 banks and $p = 2$ give 27 coefficients per equation from 250 days; more banks need shrinkage [Demirer et al. \(2018\)](#)
 - ▶ Test: compare with a LASSO VAR; unstable edges signal estimation noise, not structure

Appendix: Further Study on Quantinar

Advanced courses on quantinar.com

- ▣  Financial Risk Meter for Emerging Markets
- ▣  Dynamic Crypto Networks
- ▣  Network Centrality

Code and data

- ▣ Quantlets: github.com/danpele/MFM/tree/main/Quantlets/Ch_18

References (1/5)

- ▣ Acharya, V.V., Engle, R., Richardson, M. (2012). Capital shortfall: a new approach to ranking and regulating systemic risks. *American Economic Review*, 102(3), 59–64.
- ▣ Acharya, V.V., Pedersen, L.H., Philippon, T., Richardson, M. (2017). Measuring systemic risk. *Review of Financial Studies*, 30(1), 2–47.
- ▣ Adrian, T., Brunnermeier, M.K. (2016). CoVaR. *American Economic Review*, 106(7), 1705–1741.
- ▣ Allen, L., Bali, T.G., Tang, Y. (2012). Does systemic risk in the financial sector predict future economic downturns?. *Review of Financial Studies*, 25(10), 3000–3036.
- ▣ Ando, T., Bai, J. (2020). Quantile co-movement in financial markets: a panel quantile model with unobserved heterogeneity. *Journal of the American Statistical Association*, 115(529), 266–279.
- ▣ Ando, T., Greenwood-Nimmo, M., Shin, Y. (2022). Quantile connectedness: modeling tail behavior in the topology of financial networks. *Management Science*, 68(4), 2401–2431.
- ▣ Banulescu-Radu, D., Hurlin, C., Leymarie, J., Scaillet, O. (2021). Backtesting marginal expected shortfall and related systemic risk measures. *Management Science*, 67(9), 5730–5754.
- ▣ Barigozzi, M., Brownlees, C. (2019). NETS: network estimation for time series. *Journal of Applied Econometrics*, 34(3), 347–364.
- ▣ Baruník, J., Křehlík, T. (2018). Measuring the frequency dynamics of financial connectedness and systemic risk. *Journal of Financial Econometrics*, 16(2), 271–296.
- ▣ Basel Committee on Banking Supervision (2018). Global systemically important banks: revised assessment methodology and the higher loss absorbency requirement. Bank for International Settlements.
- ▣ Battiston, S., Mandel, A., Monasterolo, I., Schütze, F., Visentin, G. (2017). A climate stress-test of the financial system. *Nature Climate Change*, 7(4), 283–288.

References (2/5)

- ▣ Belloni, A., Chernozhukov, V. (2011). ℓ_1 -penalized quantile regression in high-dimensional sparse models. *Annals of Statistics*, 39(1), 82–130.
- ▣ Ben Amor, S., Althof, M., Härdle, W.K. (2022). Financial Risk Meter for emerging markets. *Research in International Business and Finance*, 60, 101594.
- ▣ Benjamini, Y., Hochberg, Y. (1995). Controlling the false discovery rate: a practical and powerful approach to multiple testing. *Journal of the Royal Statistical Society B*, 57(1), 289–300.
- ▣ Benoit, S., Colliard, J.-E., Hurlin, C., Pérignon, C. (2017). Where the risks lie: a survey on systemic risk. *Review of Finance*, 21(1), 109–152.
- ▣ Billio, M., Getmansky, M., Lo, A.W., Pelizzon, L. (2012). Econometric measures of connectedness and systemic risk in the finance and insurance sectors. *Journal of Financial Economics*, 104(3), 535–559.
- ▣ Bolton, P., Després, M., Pereira da Silva, L.A., Samama, F., Svartzman, R. (2020). The green swan: central banking and financial stability in the age of climate change. Bank for International Settlements.
- ▣ Brownlees, C., Engle, R.F. (2017). SRISK: a conditional capital shortfall measure of systemic risk. *Review of Financial Studies*, 30(1), 48–79.
- ▣ Castro, C., Ferrari, S. (2014). Measuring and testing for the systemically important financial institutions. *Journal of Empirical Finance*, 25, 1–14.
- ▣ Chernozhukov, V., Fernández-Val, I. (2011). Inference for extremal conditional quantile models, with an application to market and birthweight risks. *Review of Economic Studies*, 78(2), 559–589.
- ▣ Danielsson, J., James, K.R., Valenzuela, M., Zer, I. (2016). Model risk of risk models. *Journal of Financial Stability*, 23, 79–91.
- ▣ Demirer, M., Diebold, F.X., Liu, L., Yilmaz, K. (2018). Estimating global bank network connectedness. *Journal of Applied Econometrics*, 33(1), 1–15.

References (3/5)

- ▣ Diebold, F.X., Yilmaz, K. (2009). [Measuring financial asset return and volatility spillovers, with application to global equity markets.](#) *Economic Journal*, 119(534), 158–171.
- ▣ Diebold, F.X., Yilmaz, K. (2012). [Better to give than to receive: predictive directional measurement of volatility spillovers.](#) *International Journal of Forecasting*, 28(1), 57–66.
- ▣ Diebold, F.X., Yilmaz, K. (2014). [On the network topology of variance decompositions: measuring the connectedness of financial firms.](#) *Journal of Econometrics*, 182(1), 119–134.
- ▣ ECB Banking Supervision (2022). [Banks must sharpen their focus on climate risk, ECB supervisory stress test shows.](#) Press release, 8 July 2022.
- ▣ Engle, R. (2002). [Dynamic conditional correlation: a simple class of multivariate generalized autoregressive conditional heteroskedasticity models.](#) *Journal of Business & Economic Statistics*, 20(3), 339–350.
- ▣ Engle, R., Jondeau, E., Rockinger, M. (2015). [Systemic risk in Europe.](#) *Review of Finance*, 19(1), 145–190.
- ▣ FDIC (2023). [FDIC creates a Deposit Insurance National Bank of Santa Clara to protect insured depositors of Silicon Valley Bank, Santa Clara, California.](#) Press release, 10 March 2023.
- ▣ Federal Reserve Board (2023). [Federal Reserve Board announces it will make available additional funding to eligible depository institutions.](#) Press release, 12 March 2023.
- ▣ Franke, J., Härdle, W.K., Hafner, C.M. (2019). [Statistics of Financial Markets: An Introduction](#), 5th ed. Springer.
- ▣ Giglio, S., Kelly, B., Pruitt, S. (2016). [Systemic risk and the macroeconomy: an empirical evaluation.](#) *Journal of Financial Economics*, 119(3), 457–471.
- ▣ Girardi, G., Ergün, A.T. (2013). [Systemic risk measurement: multivariate GARCH estimation of CoVaR.](#) *Journal of Banking & Finance*, 37(8), 3169–3180.

References (4/5)

- ▣ Glostén, L.R., Jagannathan, R., Runkle, D.E. (1993). On the relation between the expected value and the volatility of the nominal excess return on stocks. *Journal of Finance*, 48(5), 1779–1801.
- ▣ Gouriéroux, C., Laurent, J.P., Scaillet, O. (2000). Sensitivity analysis of values at risk. *Journal of Empirical Finance*, 7(3–4), 225–245.
- ▣ Härdle, W.K., Chen, C.Y.-H., Overbeck, L. (Eds.) (2017). *Applied Quantitative Finance*, 3rd ed. Springer.
- ▣ Härdle, W.K., Wang, W., Yu, L. (2016). TENET: Tail-Event driven NETWORK risk. *Journal of Econometrics*, 192(2), 499–513.
- ▣ Holm, S. (1979). A simple sequentially rejective multiple test procedure. *Scandinavian Journal of Statistics*, 6(2), 65–70.
- ▣ Koenker, R., Bassett, G. (1978). Regression quantiles. *Econometrica*, 46(1), 33–50.
- ▣ Koenker, R., Bassett, G. (1982). Tests of linear hypotheses and ℓ_1 estimation. *Econometrica*, 50(6), 1577–1583.
- ▣ Koop, G., Pesaran, M.H., Potter, S.M. (1996). Impulse response analysis in nonlinear multivariate models. *Journal of Econometrics*, 74(1), 119–147.
- ▣ Mihoci, A., Althof, M., Chen, C.Y.-H., Härdle, W.K. (2020). FRM Financial Risk Meter. In *The Econometrics of Networks*, Advances in Econometrics 42, Emerald, 335–368.
- ▣ Nucera, F., Schwaab, B., Koopman, S.J., Lucas, A. (2016). The information in systemic risk rankings. *Journal of Empirical Finance*, 38, 461–475.
- ▣ Parkinson, M. (1980). The extreme value method for estimating the variance of the rate of return. *Journal of Business*, 53(1), 61–65.

References (5/5)

- ▣ Pele, D.T., Lazar, E., Dufour, A. (2017). Information entropy and measures of market risk. *Entropy*, 19(5), 226.
- ▣ Pesaran, H.H., Shin, Y. (1998). Generalized impulse response analysis in linear multivariate models. *Economics Letters*, 58(1), 17–29.
- ▣ Ren, R., Althof, M., Härdle, W.K. (2022). Financial Risk Meter for cryptocurrencies and tail risk network-based portfolio construction. *Singapore Economic Review*, 70(4), 901–927.
- ▣ Ren, R., Lu, M., Li, Y., Härdle, W.K. (2022). Financial Risk Meter FRM based on expectiles. *Journal of Multivariate Analysis*, 189, 104881.
- ▣ Scaillet, O. (2004). Nonparametric estimation and sensitivity analysis of expected shortfall. *Mathematical Finance*, 14(1), 115–129.
- ▣ Shi, S., Hurn, S., Phillips, P.C.B. (2020). Causal change detection in possibly integrated systems: revisiting the money–income relationship. *Journal of Financial Econometrics*, 18(1), 158–180.
- ▣ Toda, H.Y., Yamamoto, T. (1995). Statistical inference in vector autoregressions with possibly integrated processes. *Journal of Econometrics*, 66(1–2), 225–250.
- ▣ Toma, F.-M., Dima, D.M., Grecu, R., Pele, D.T., Lessmann, S. (2026). Energy Financial Risk Meter: systemic risk in the EU energy sector. SSRN working paper.
- ▣ Wang, R., Althof, M., Härdle, W.K. (2023). A financial risk meter for China. *Emerging Markets Review*, 56, 101052.